

ECG examples - PREPROCESSED (z-scored), no reconstruction

52 example beats, grouped by MIT-BIH annotation type

Threshold (model decision): 0.0434

Distribution:

N Normal sinus beat	10
F Fusion of ventricular and normal	9
V Premature ventricular contraction (PVC)	8
L Left bundle branch block	8
A Atrial premature beat (APC)	6
R Right bundle branch block	5
/ Paced beat	4
f Fusion of paced and normal	2

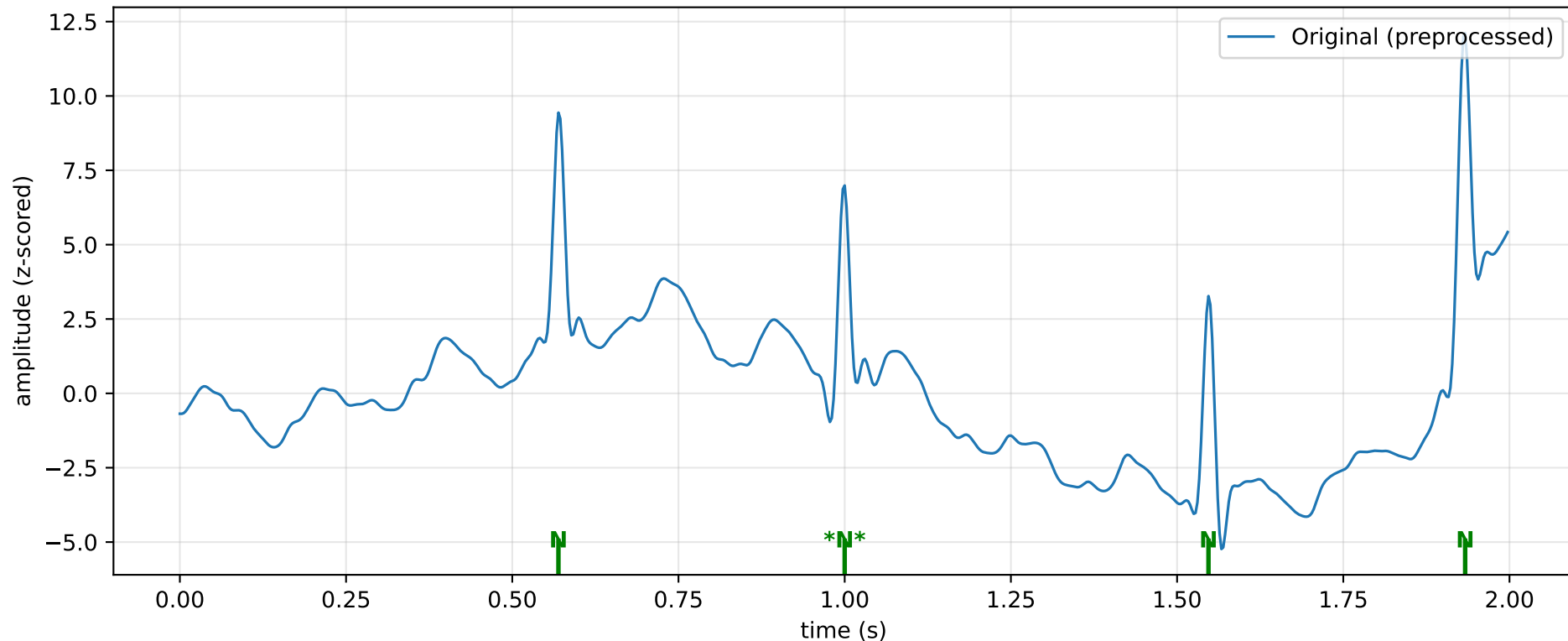
Labels on each page:

Symbol	symbol of the centre beat (the one that labels the window)
Type	plain-language meaning
Source	MIT-BIH record + sample index (reproducible)
In window	ALL annotated beats inside this 2-second window, listed as time-from-centre:symbol (e.g. -380ms:N *0ms:V* +420ms:N) Asterisks mark the centre beat - the one whose label is the window label. Other beats also affect the residual but their labels are NOT used. This is the per-beat-label limitation.
Residual	model's reconstruction error on the window
Threshold	the F1-optimal value 0.0434
Decision	model's NORMAL vs ANOMALY call
Outcome	TP, FN, TN, FP

On the plot itself: short vertical ticks at the bottom mark every annotated R-peak in the window. Green = NORMAL beat, magenta = ABNORMAL beat.
The centre beat's symbol is shown in *asterisks* in the textual list.

Section: N - Normal sinus beat

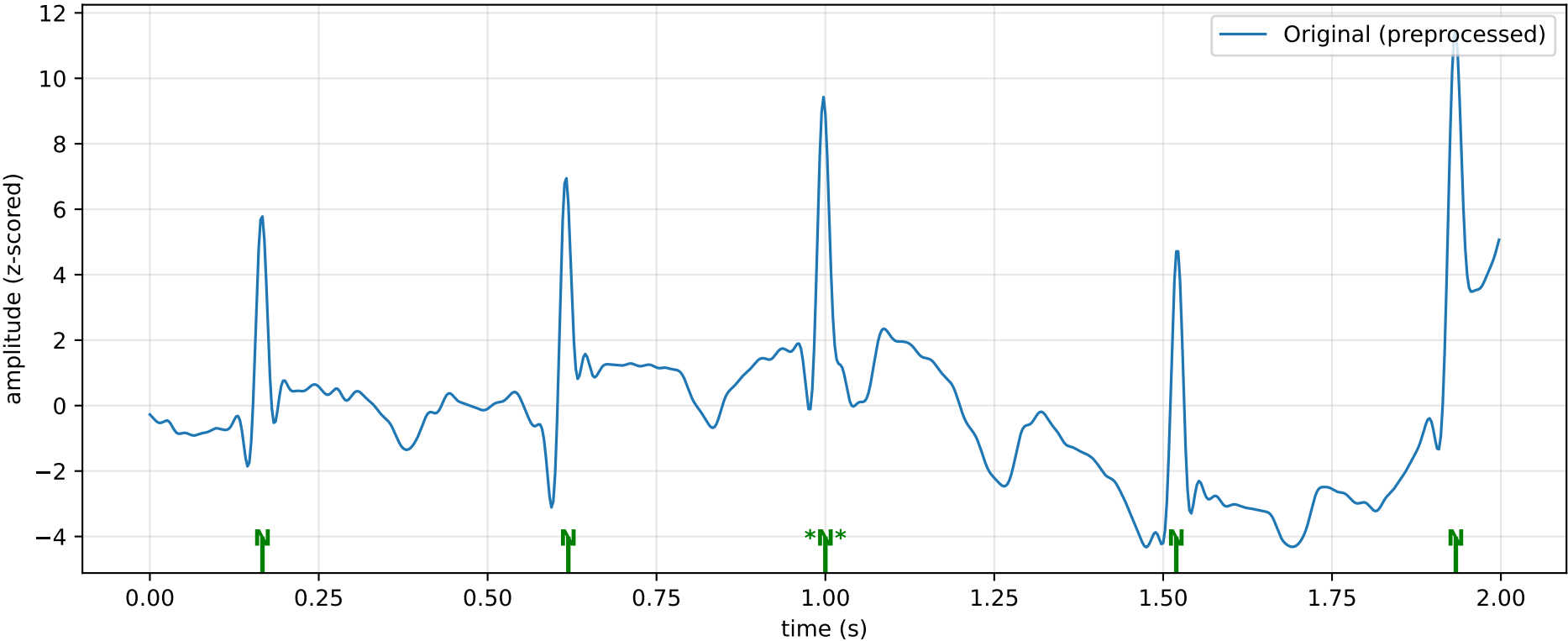
FP: Normal sinus beat (N)



Symbol	N (centre beat - the one that labels the window)
Type	Normal sinus beat
Source	record 222, sample 615454
In window	-431ms:N +0ms:*N* +547ms:N +933ms:N
Residual	1.0414 Threshold 0.0434
Decision	ANOMALY (truth: NORMAL)
Outcome	FP mismatch

ANOMALY

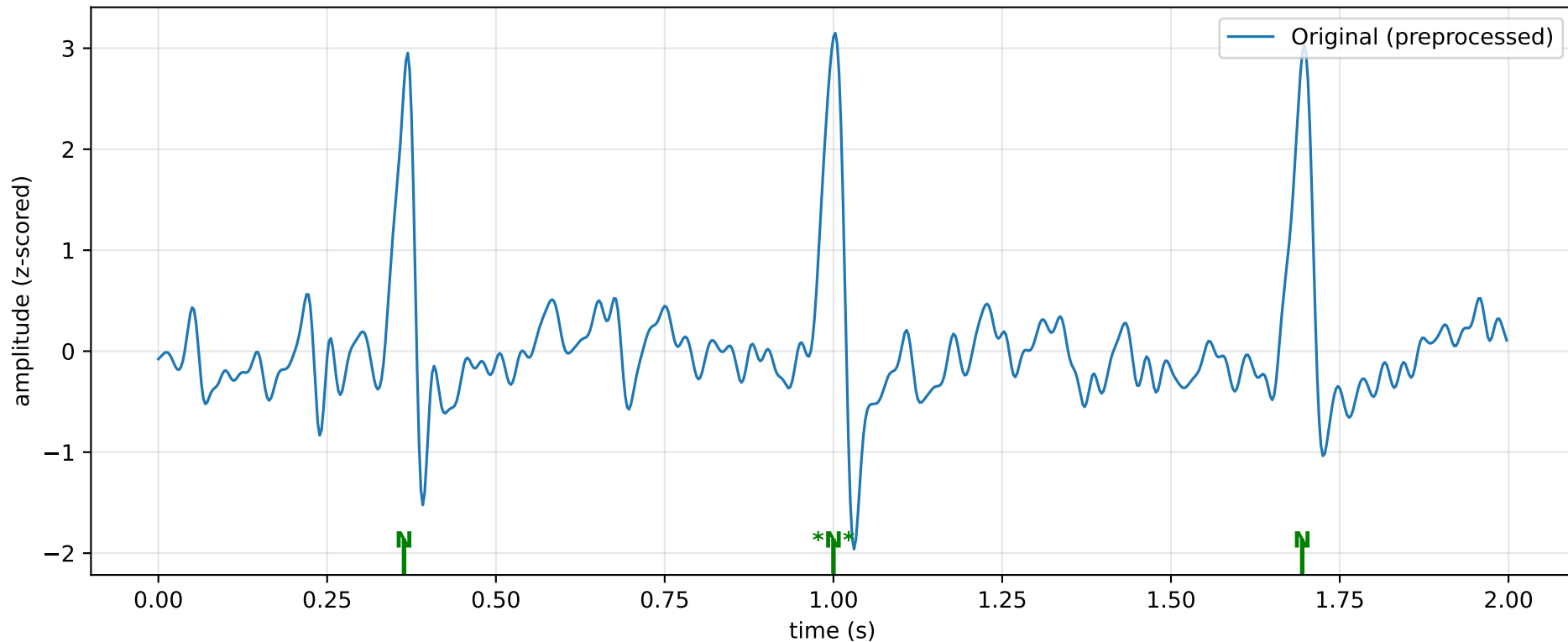
FP: Normal sinus beat (N)



Symbol	N (centre beat - the one that labels the window)				
Type	Normal sinus beat				
Source	record 222, sample 333416				
In window	-833ms:N	-381ms:N	+0ms:*N*	+519ms:N	+933ms:N
Residual	1.0266	Threshold 0.0434			
Decision	ANOMALY (truth: NORMAL)				
Outcome	FP mismatch				

ANOMALY

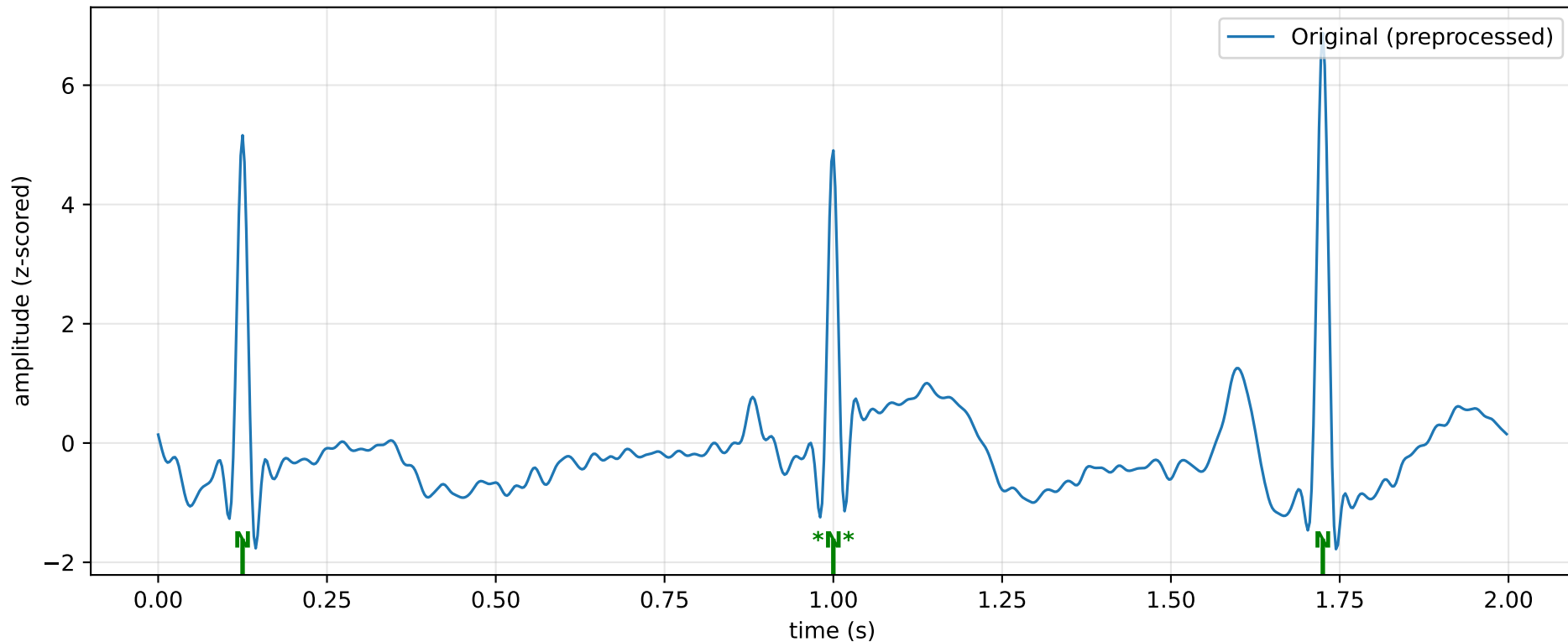
TN: Normal sinus beat (N)



Symbol	N (centre beat - the one that labels the window)
Type	Normal sinus beat
Source	record 200, sample 149157
In window	-636ms:N +0ms:*N* +694ms:N
Residual	0.0434 Threshold 0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN agree

NORMAL

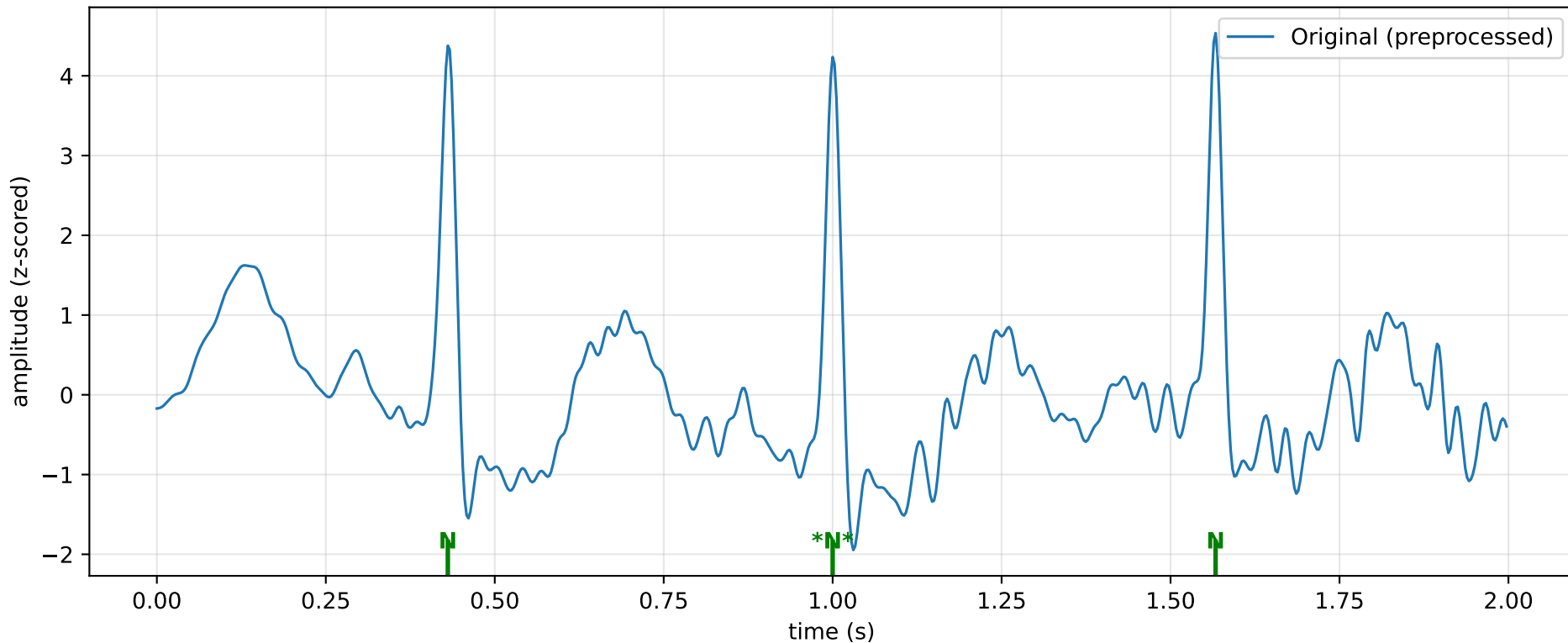
TN: Normal sinus beat (N)



Symbol	N (centre beat - the one that labels the window)
Type	Normal sinus beat
Source	record 222, sample 37058
In window	-875ms:N +0ms:*N* +725ms:N
Residual	0.0434 Threshold 0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN agree

NORMAL

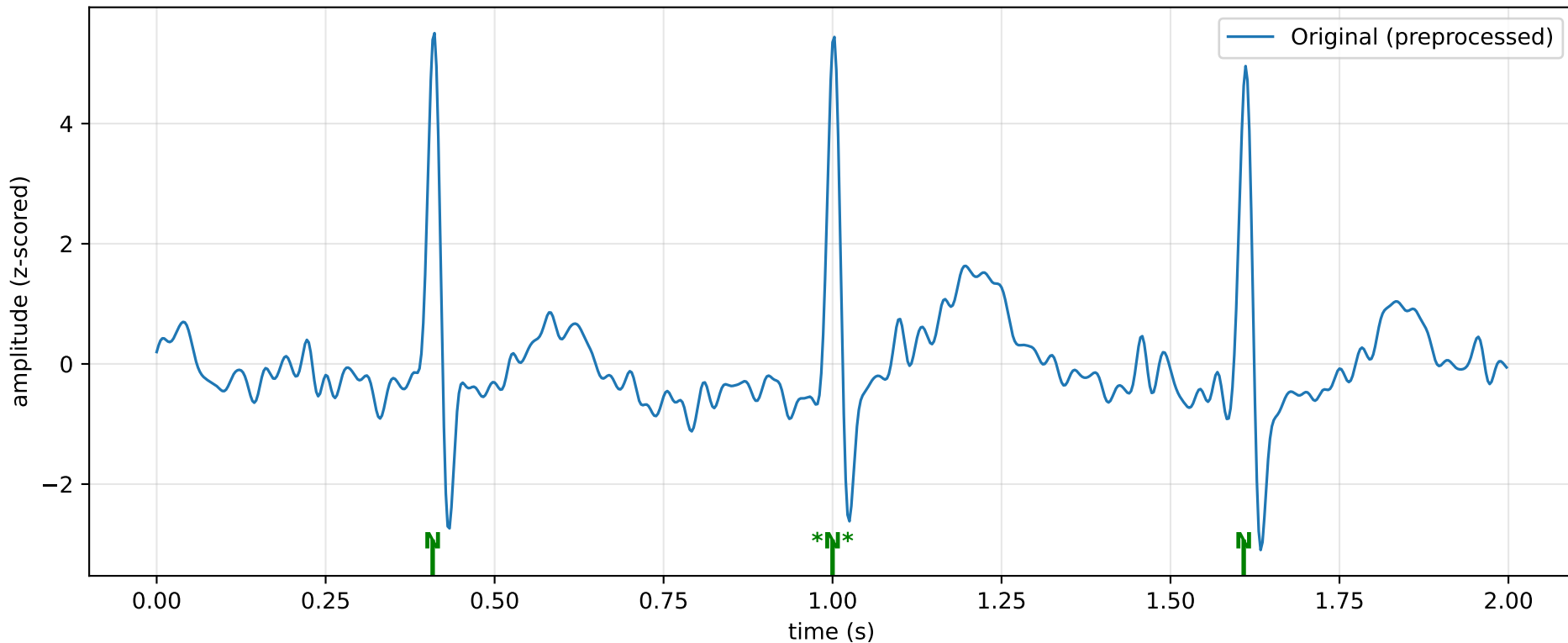
TN: Normal sinus beat (N)



Symbol	N (centre beat - the one that labels the window)
Type	Normal sinus beat
Source	record 213, sample 348145
In window	-569ms:N +0ms:*N* +567ms:N
Residual	0.0434 Threshold 0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN agree

NORMAL

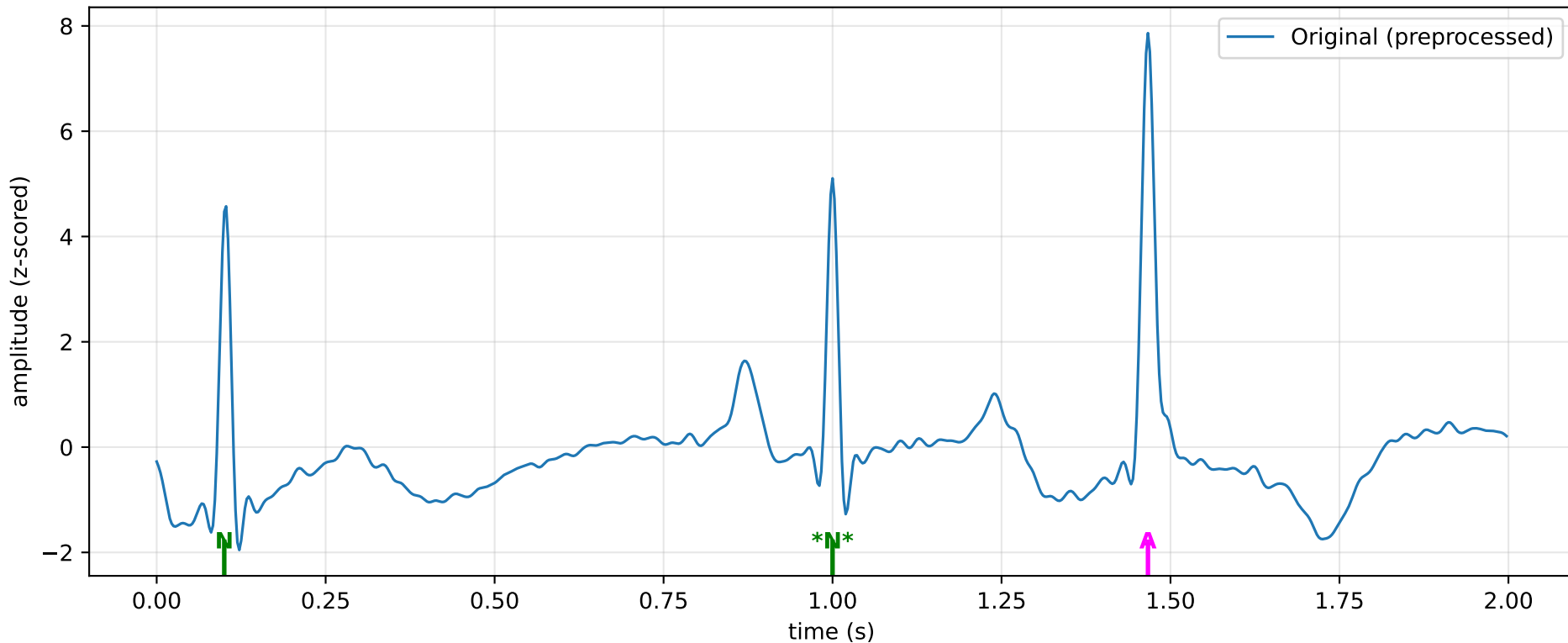
TN: Normal sinus beat (N)



Symbol	N (centre beat - the one that labels the window)
Type	Normal sinus beat
Source	record 209, sample 593746
In window	-592ms:N +0ms:*N* +608ms:N
Residual	0.0433 Threshold 0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN agree

NORMAL

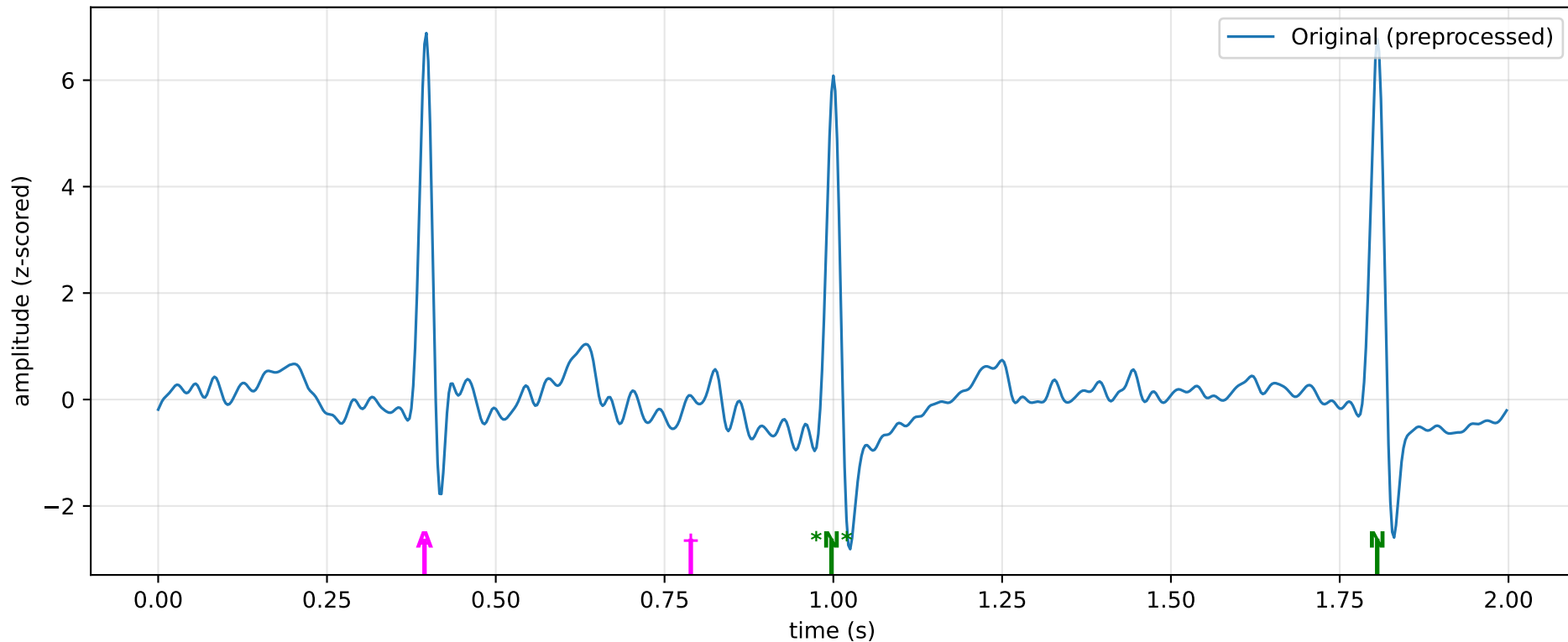
TN: Normal sinus beat (N)



Symbol	N (centre beat - the one that labels the window)
Type	Normal sinus beat
Source	record 222, sample 309870
In window	-900ms:N +0ms:*N* +467ms:A
Residual	0.0433 Threshold 0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN agree

NORMAL

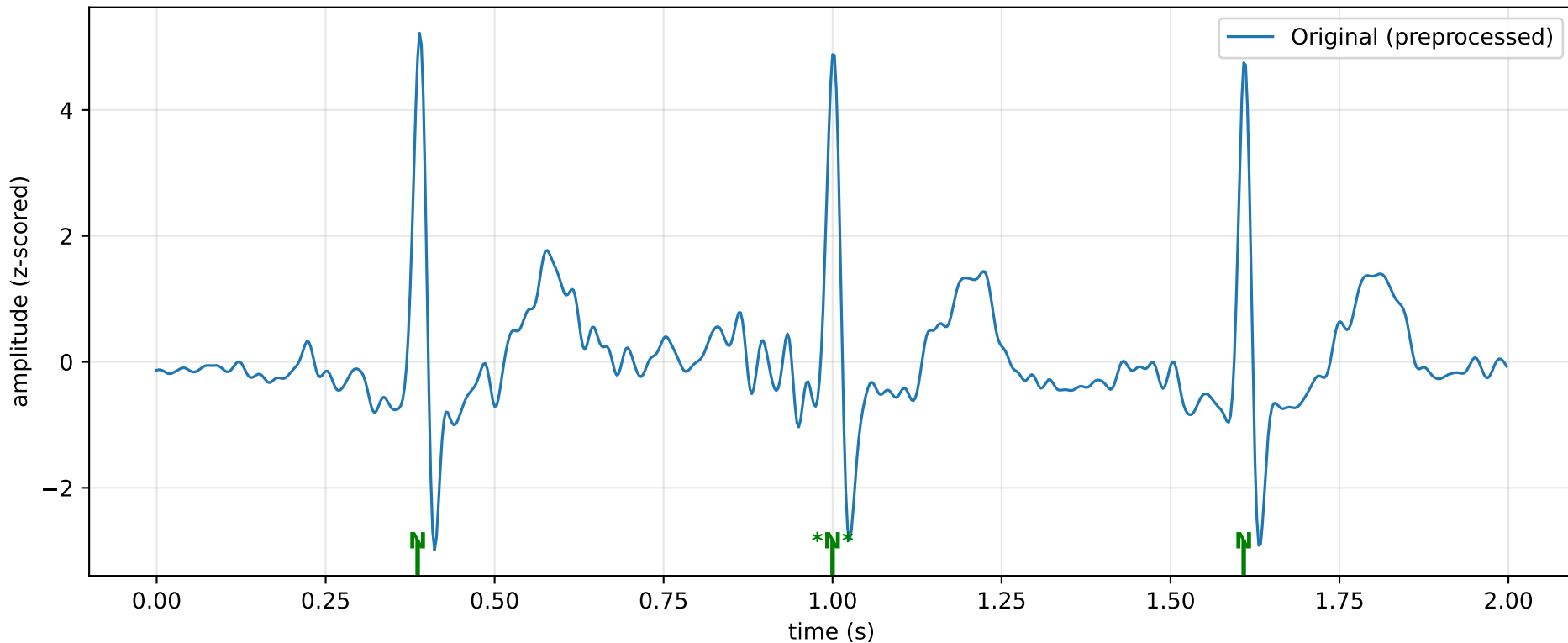
TN: Normal sinus beat (N)



Symbol	N (centre beat - the one that labels the window)
Type	Normal sinus beat
Source	record 209, sample 551919
In window	-606ms:A -211ms:+ -3ms:*N* +806ms:N
Residual	0.0433 Threshold 0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN agree

NORMAL

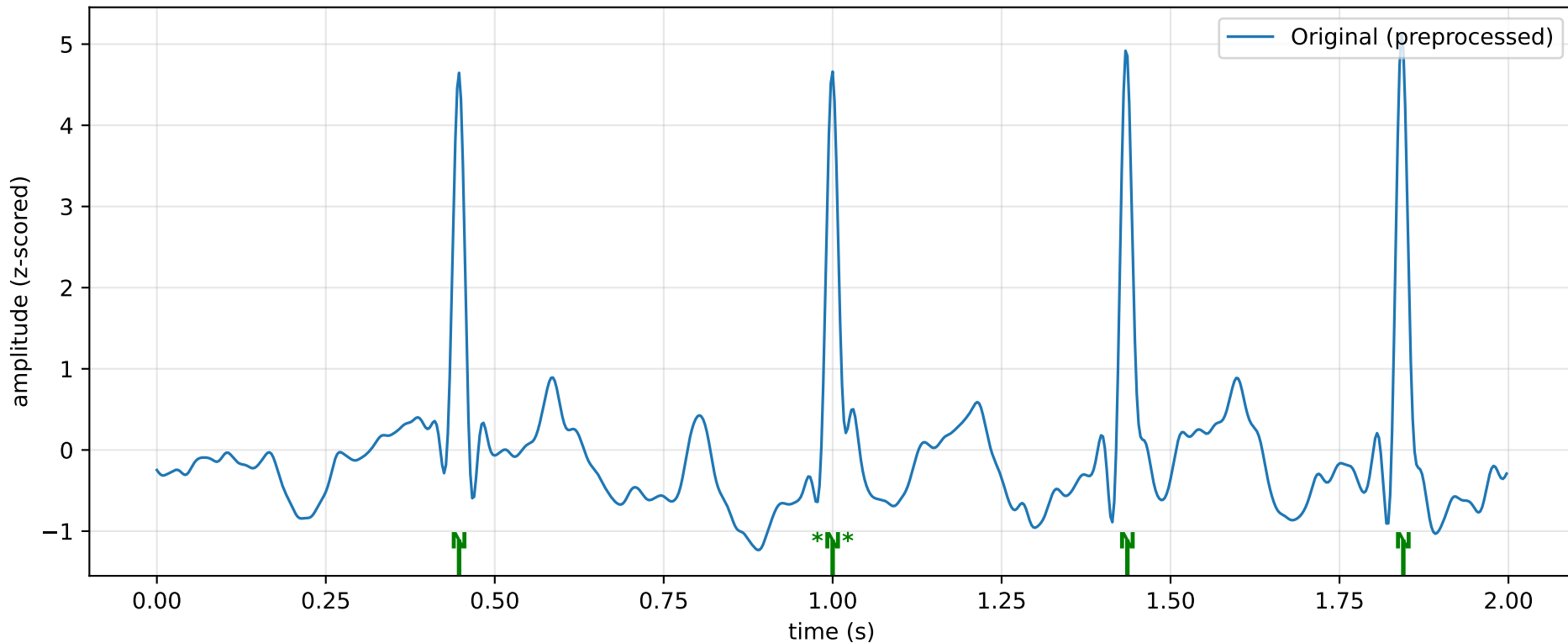
TN: Normal sinus beat (N)



Symbol	N (centre beat - the one that labels the window)
Type	Normal sinus beat
Source	record 209, sample 234212
In window	-614ms:N +0ms:*N* +608ms:N
Residual	0.0433 Threshold 0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN agree

NORMAL

TN: Normal sinus beat (N)

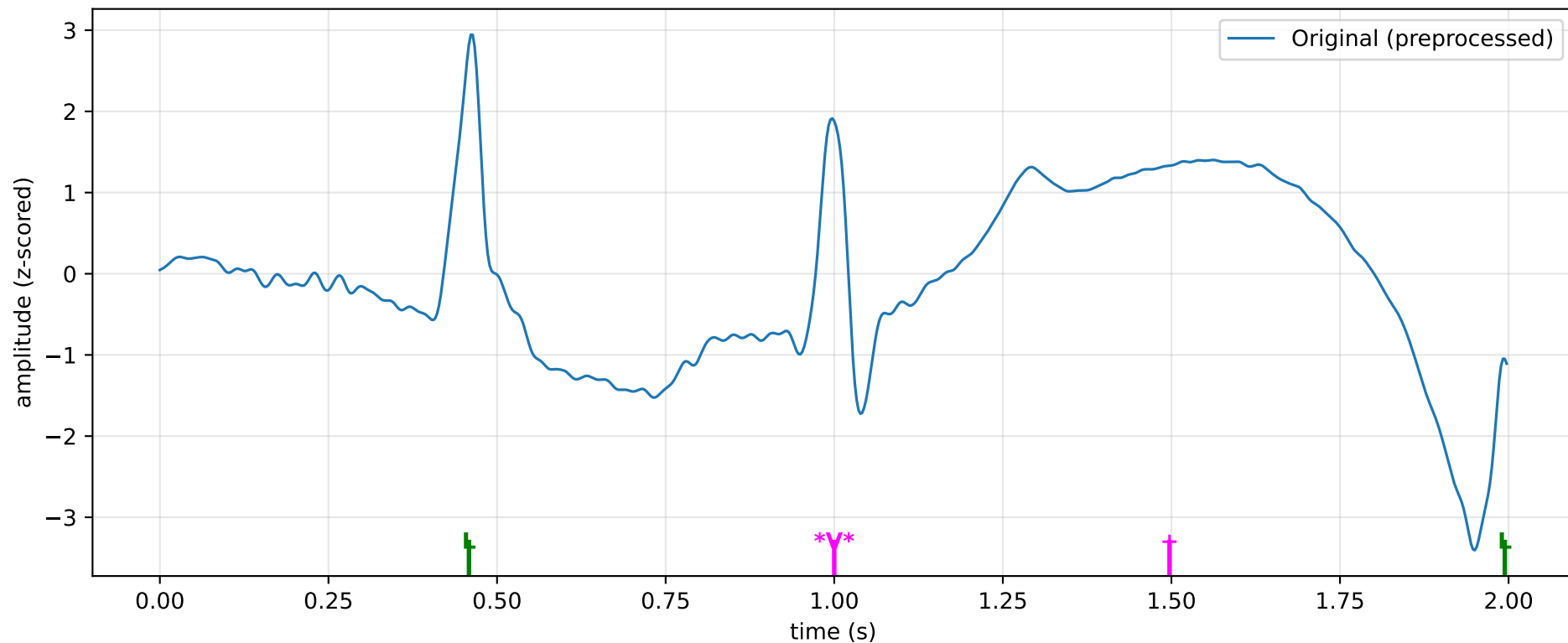


Symbol	N (centre beat - the one that labels the window)
Type	Normal sinus beat
Source	record 222, sample 580580
In window	-553ms:N +0ms:*N* +436ms:N +844ms:N
Residual	0.0433 Threshold 0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN agree

NORMAL

Section: V - Premature ventricular contraction (PVC)

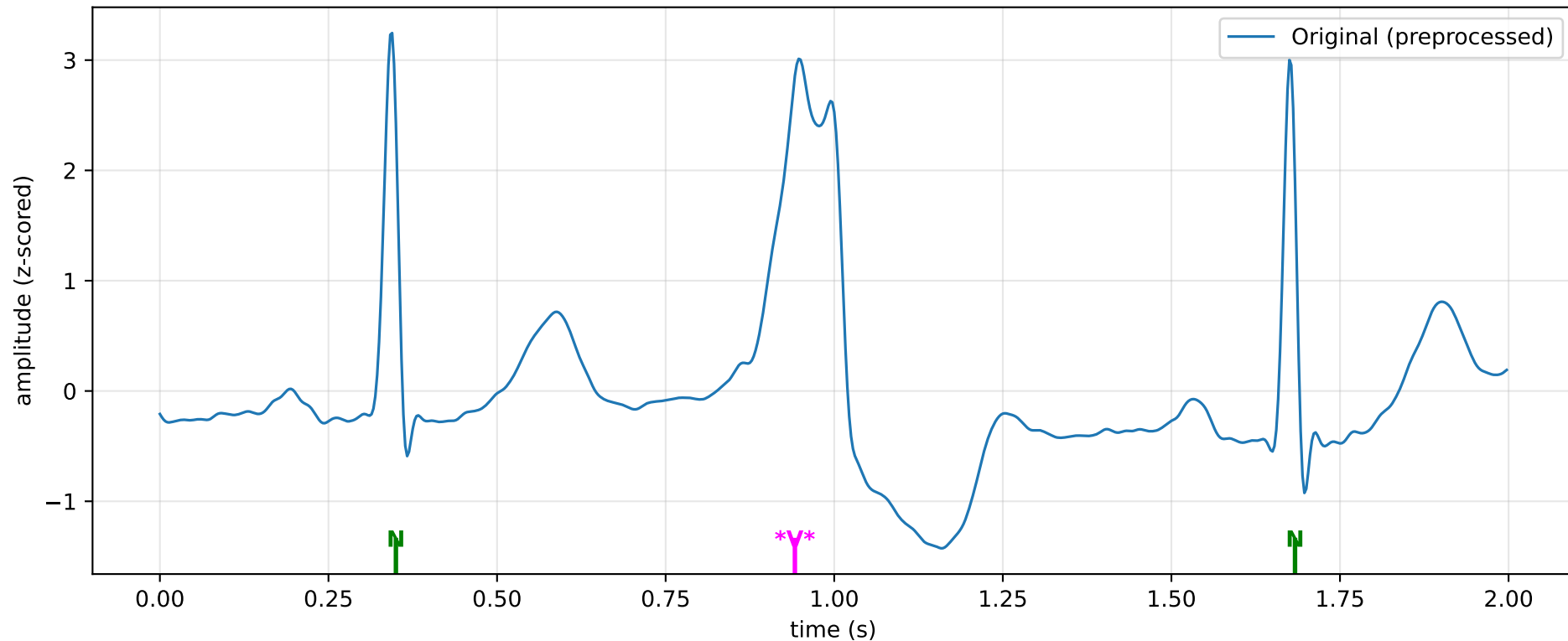
FN: Premature ventricular contraction (PVC) (V)



Symbol	V (centre beat - the one that labels the window)
Type	Premature ventricular contraction (PVC)
Source	record 214, sample 31486
In window	-542ms:L +0ms:*V* +497ms:+ +994ms:L
Residual	0.0427 Threshold 0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN mismatch

NORMAL

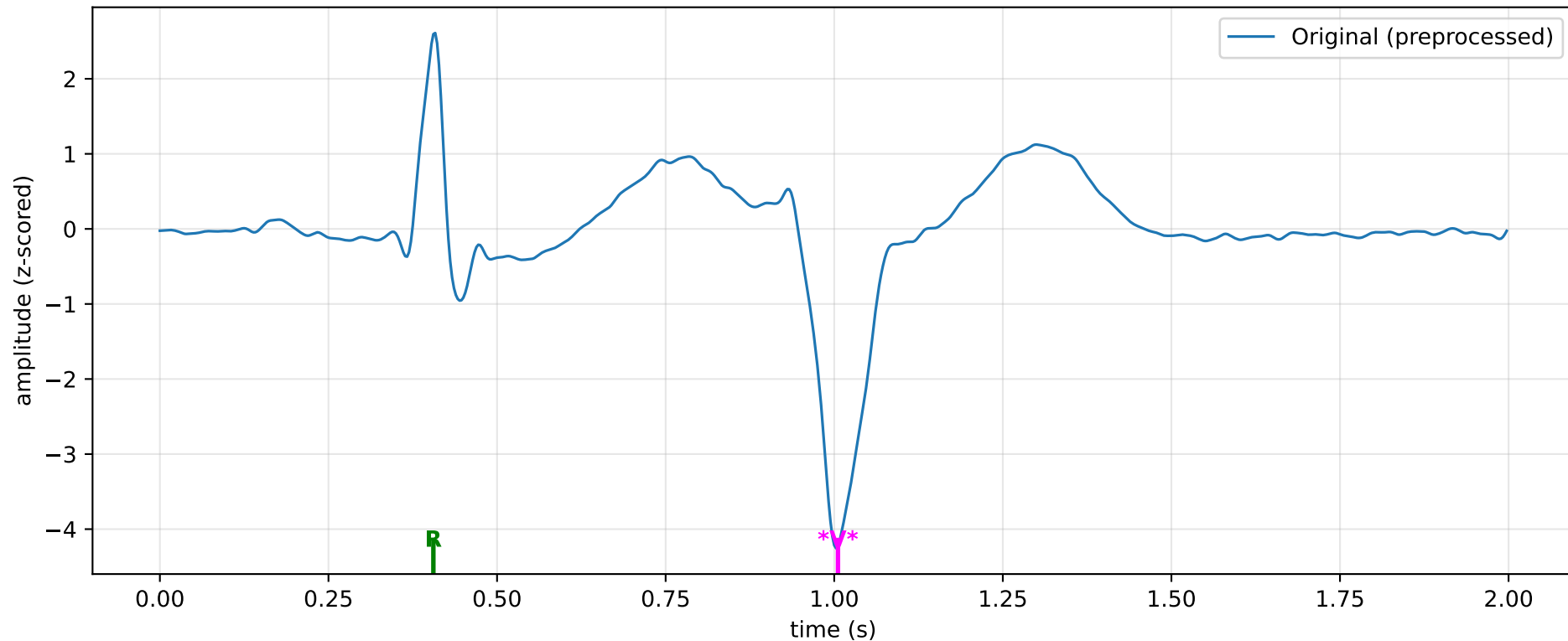
FN: Premature ventricular contraction (PVC) (V)



Symbol	V (centre beat - the one that labels the window)
Type	Premature ventricular contraction (PVC)
Source	record 208, sample 561101
In window	-650ms:N -58ms:*V* +683ms:N
Residual	0.0423 Threshold 0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN mismatch

NORMAL

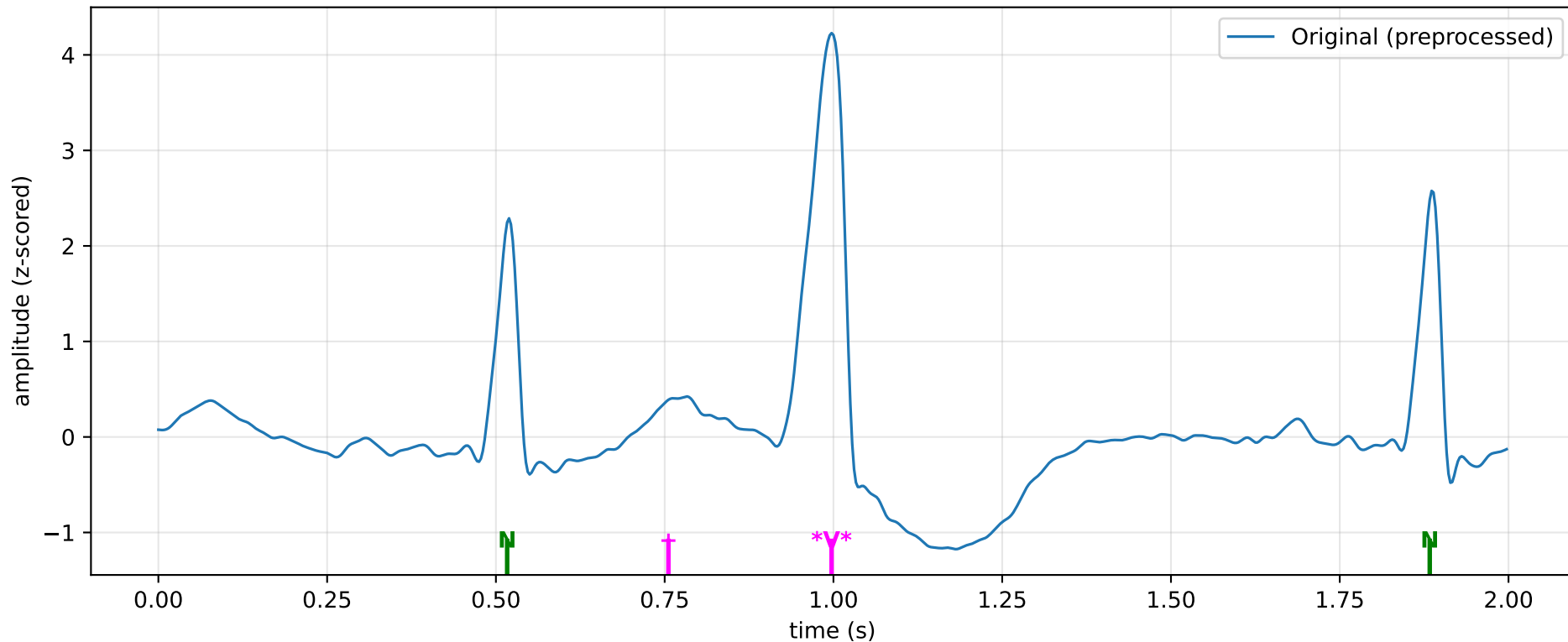
FN: Premature ventricular contraction (PVC) (V)



Symbol	V (centre beat - the one that labels the window)
Type	Premature ventricular contraction (PVC)
Source	record 207, sample 48883
In window	-594ms:R +6ms:*V*
Residual	0.0422 Threshold 0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN mismatch

NORMAL

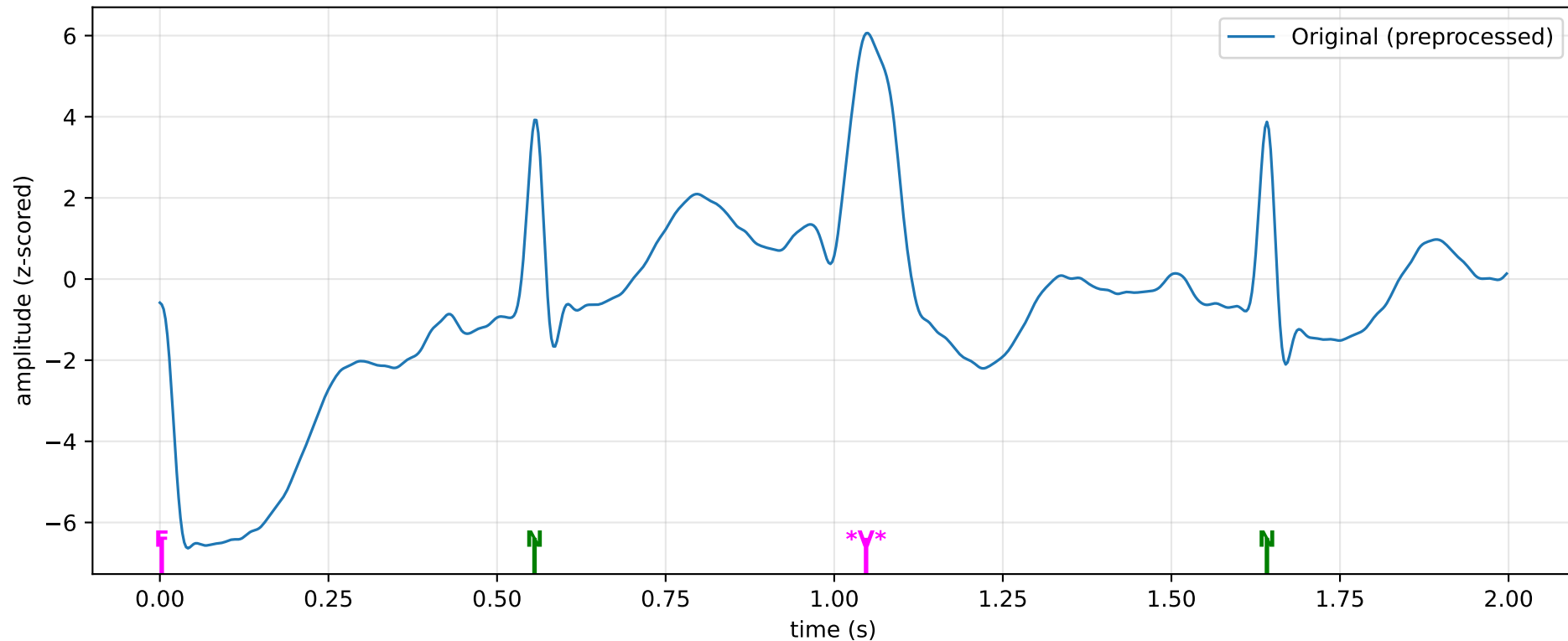
FN: Premature ventricular contraction (PVC) (V)



Symbol	V (centre beat - the one that labels the window)
Type	Premature ventricular contraction (PVC)
Source	record 200, sample 266894
In window	-483ms:N -244ms:+ -3ms:*V* +883ms:N
Residual	0.0420 Threshold 0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN mismatch

NORMAL

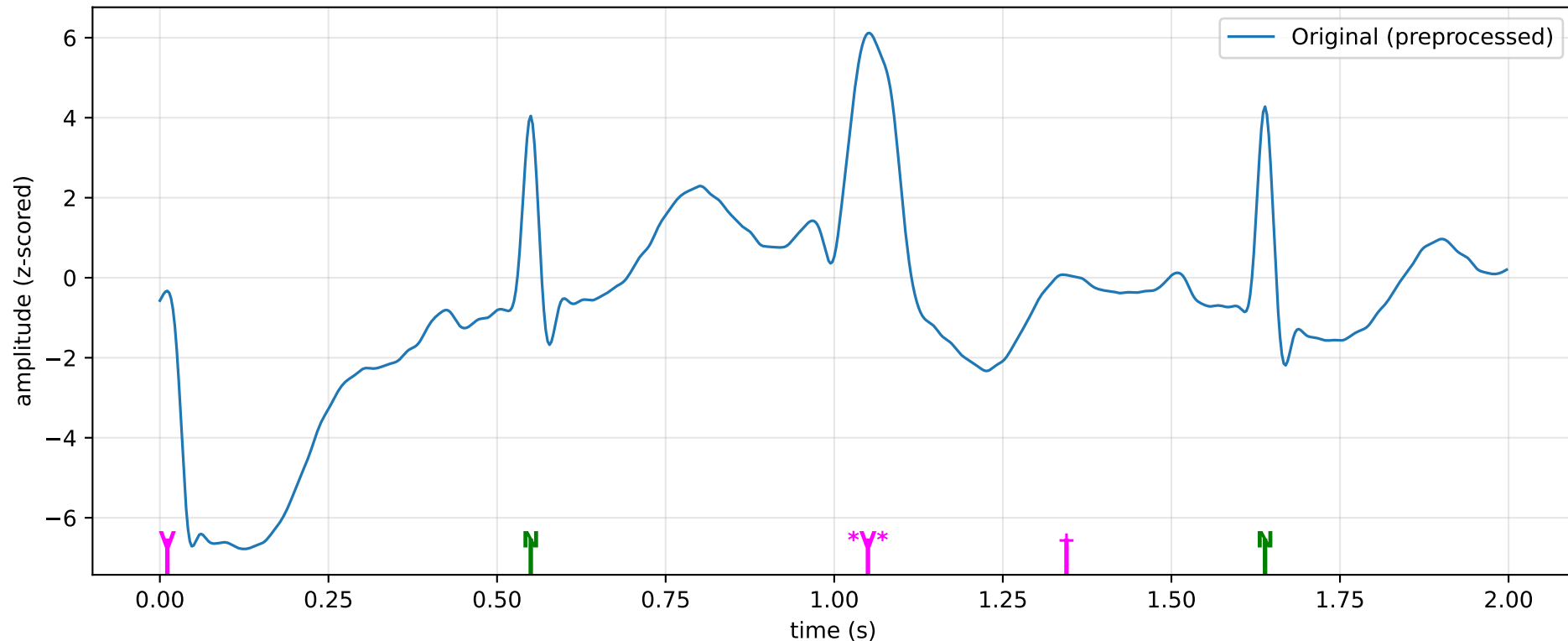
TP: Premature ventricular contraction (PVC) (V)



Symbol	V (centre beat - the one that labels the window)			
Type	Premature ventricular contraction (PVC)			
Source	record 213, sample 57428			
In window	-997ms:F	-444ms:N	+47ms:*V*	+642ms:N
Residual	1.2749	Threshold 0.0434		
Decision	ANOMALY (truth: ANOMALY)			
Outcome	TP agree			

ANOMALY

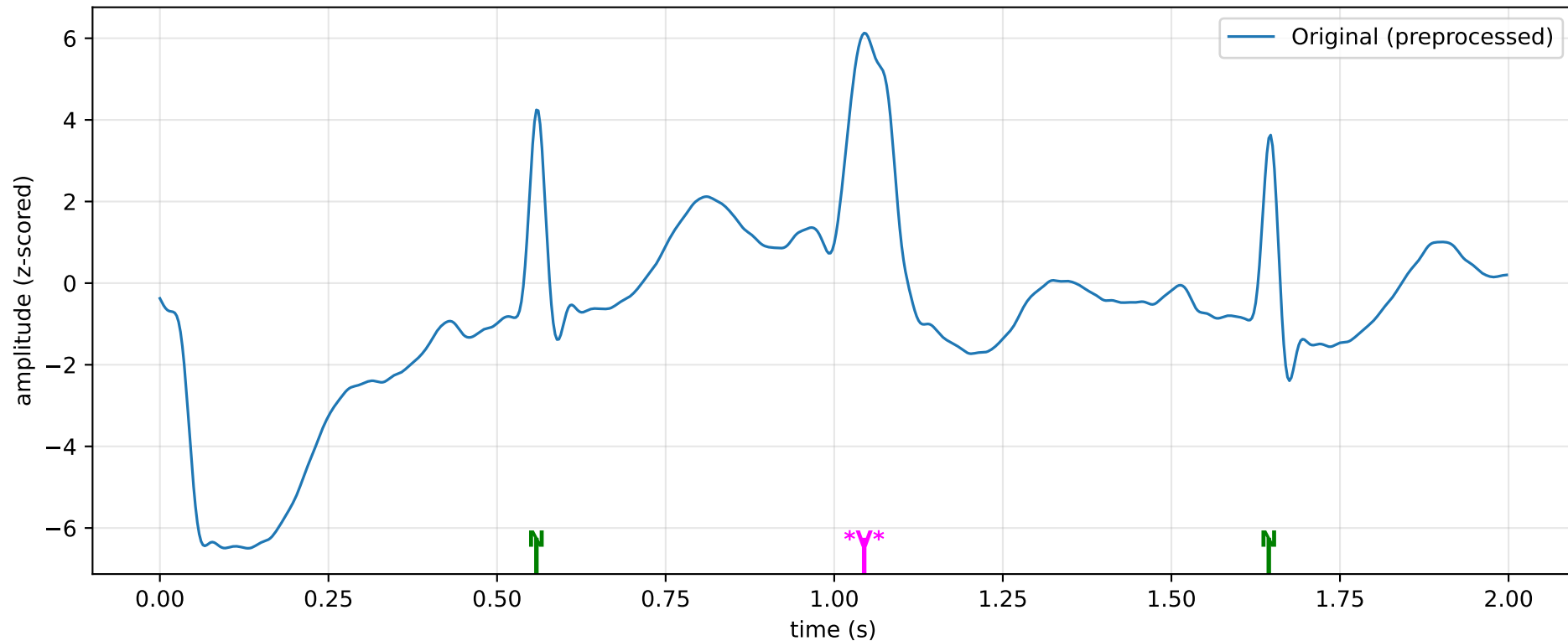
TP: Premature ventricular contraction (PVC) (V)



Symbol	V (centre beat - the one that labels the window)
Type	Premature ventricular contraction (PVC)
Source	record 213, sample 96670
In window	-989ms:V -450ms:N +50ms:*V* +344ms:+ +639ms:N
Residual	1.2723 Threshold 0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP agree

ANOMALY

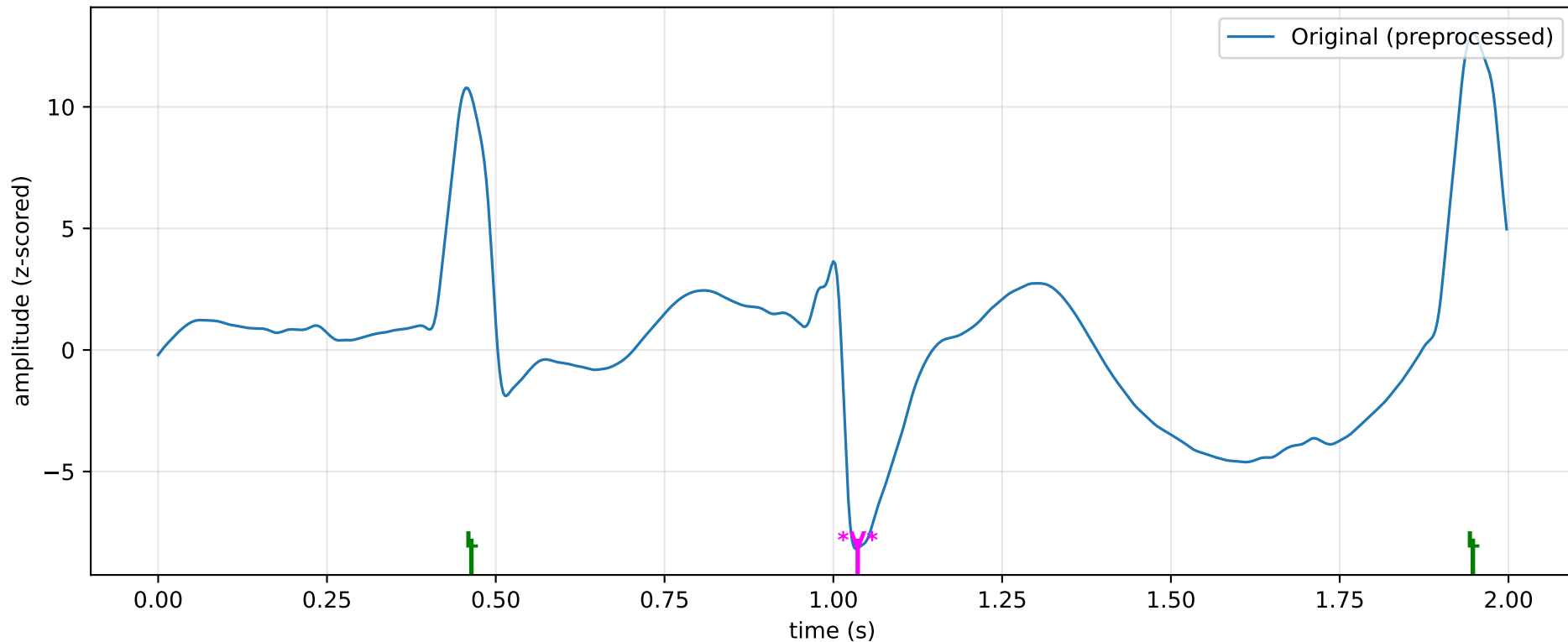
TP: Premature ventricular contraction (PVC) (V)



Symbol	V (centre beat - the one that labels the window)
Type	Premature ventricular contraction (PVC)
Source	record 213, sample 167531
In window	-442ms:N +44ms:*V* +644ms:N
Residual	1.2521 Threshold 0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP agree

ANOMALY

TP: Premature ventricular contraction (PVC) (V)

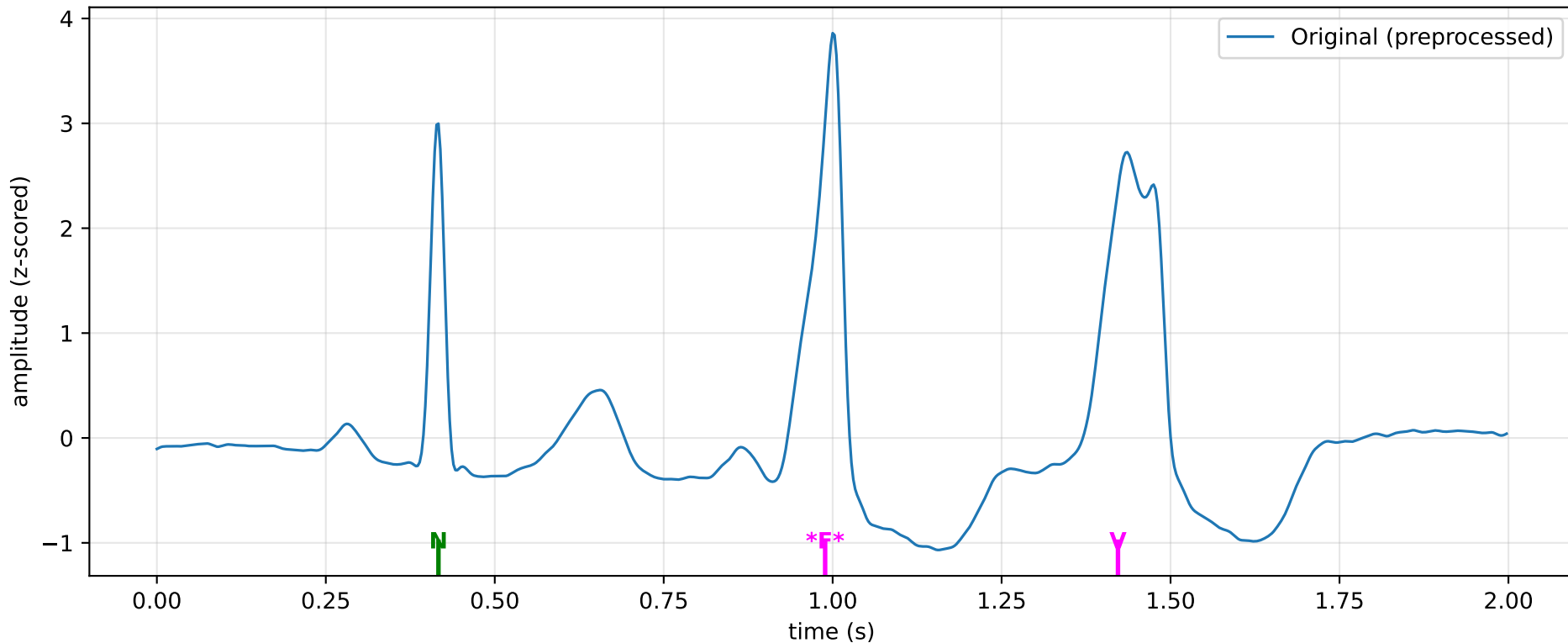


Symbol	V (centre beat - the one that labels the window)
Type	Premature ventricular contraction (PVC)
Source	record 109, sample 603535
In window	-536ms:L +36ms:*V* +947ms:L
Residual	1.1877 Threshold 0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP agree

ANOMALY

Section: F - Fusion of ventricular and normal

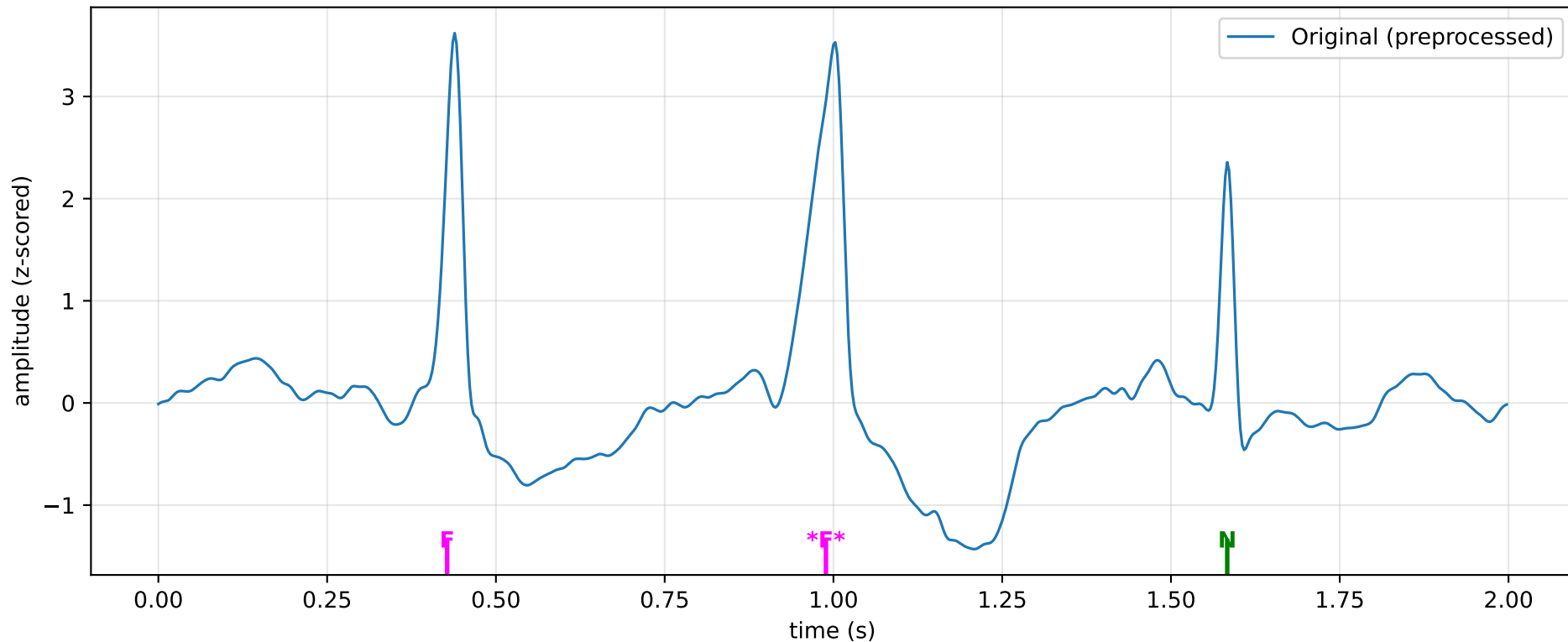
FN: Fusion of ventricular and normal (F)



Symbol	F (centre beat - the one that labels the window)
Type	Fusion of ventricular and normal
Source	record 208, sample 203236
In window	-583ms:N -11ms:*F* +422ms:V
Residual	0.0434 Threshold 0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN mismatch

NORMAL

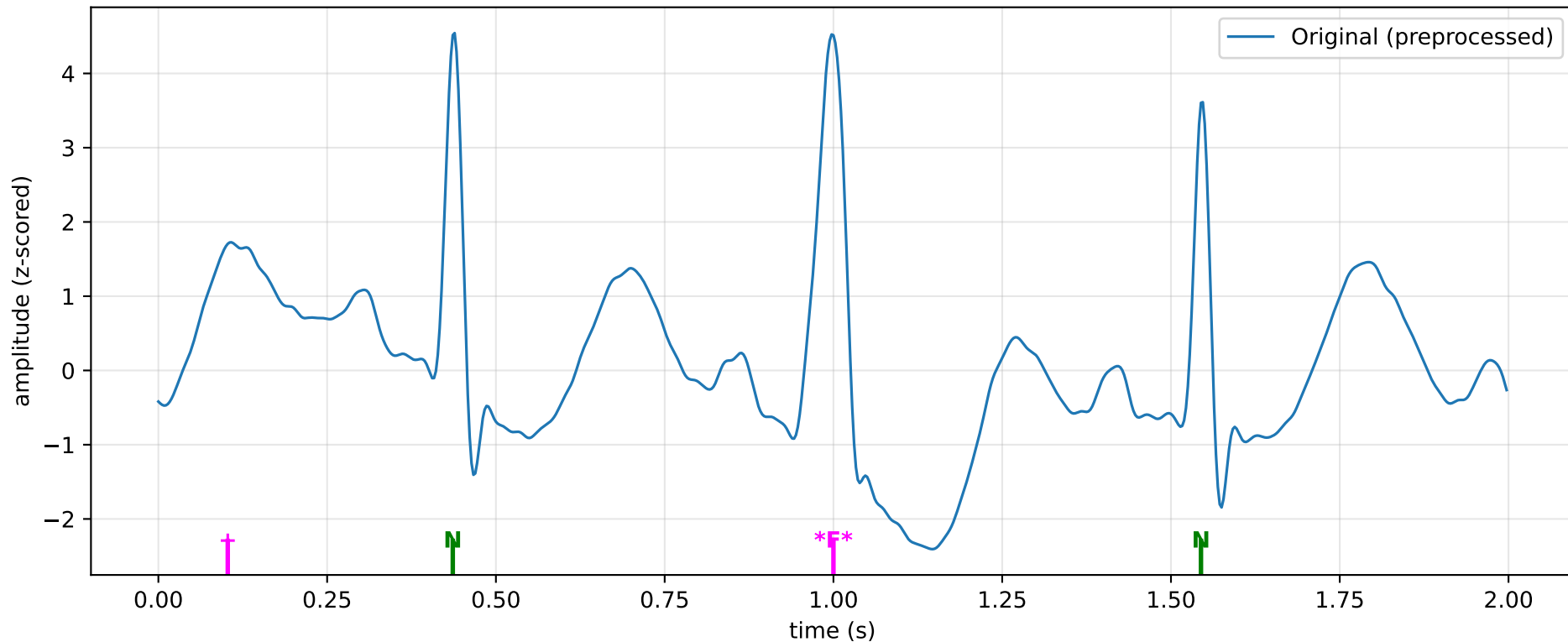
FN: Fusion of ventricular and normal (F)



Symbol	F (centre beat - the one that labels the window)
Type	Fusion of ventricular and normal
Source	record 208, sample 429272
In window	-572ms:F -11ms:*F* +583ms:N
Residual	0.0434 Threshold 0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN mismatch

NORMAL

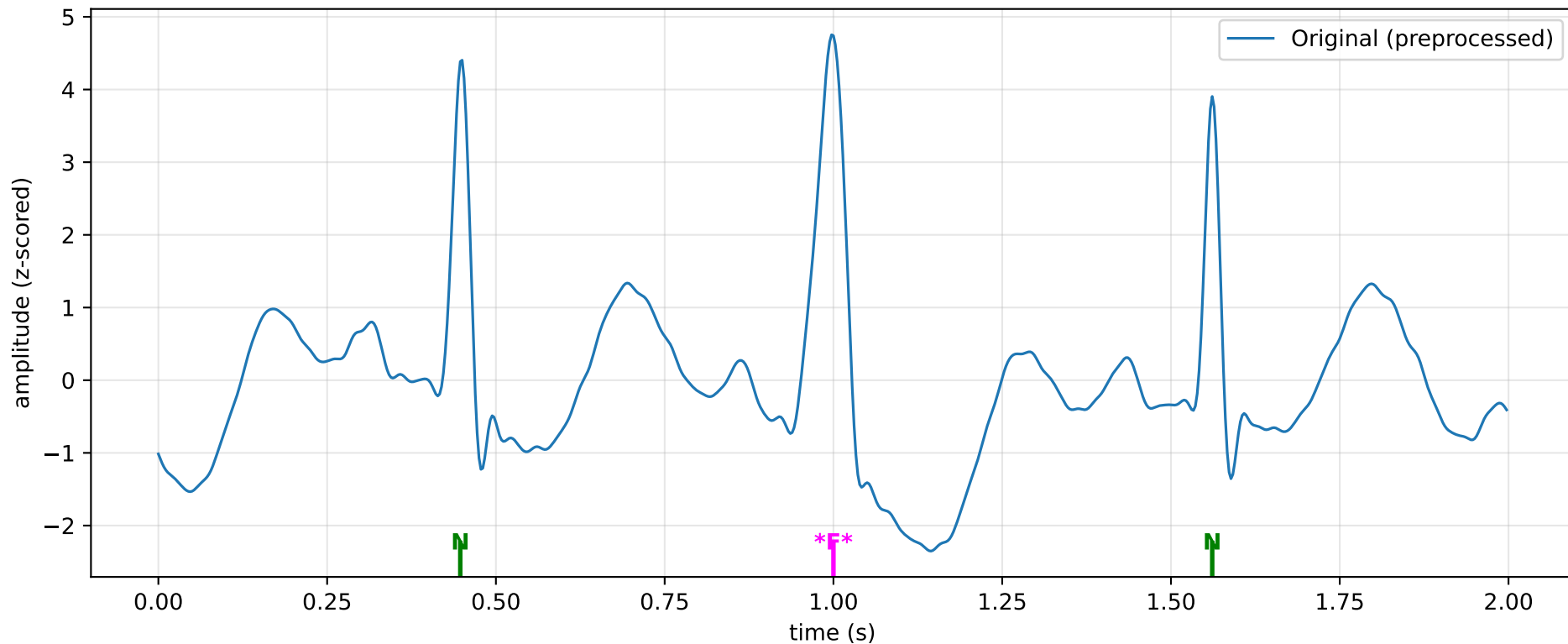
FN: Fusion of ventricular and normal (F)



Symbol	F (centre beat - the one that labels the window)
Type	Fusion of ventricular and normal
Source	record 213, sample 377920
In window	-897ms:+ -564ms:N +0ms:*F* +544ms:N
Residual	0.0433 Threshold 0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN mismatch

NORMAL

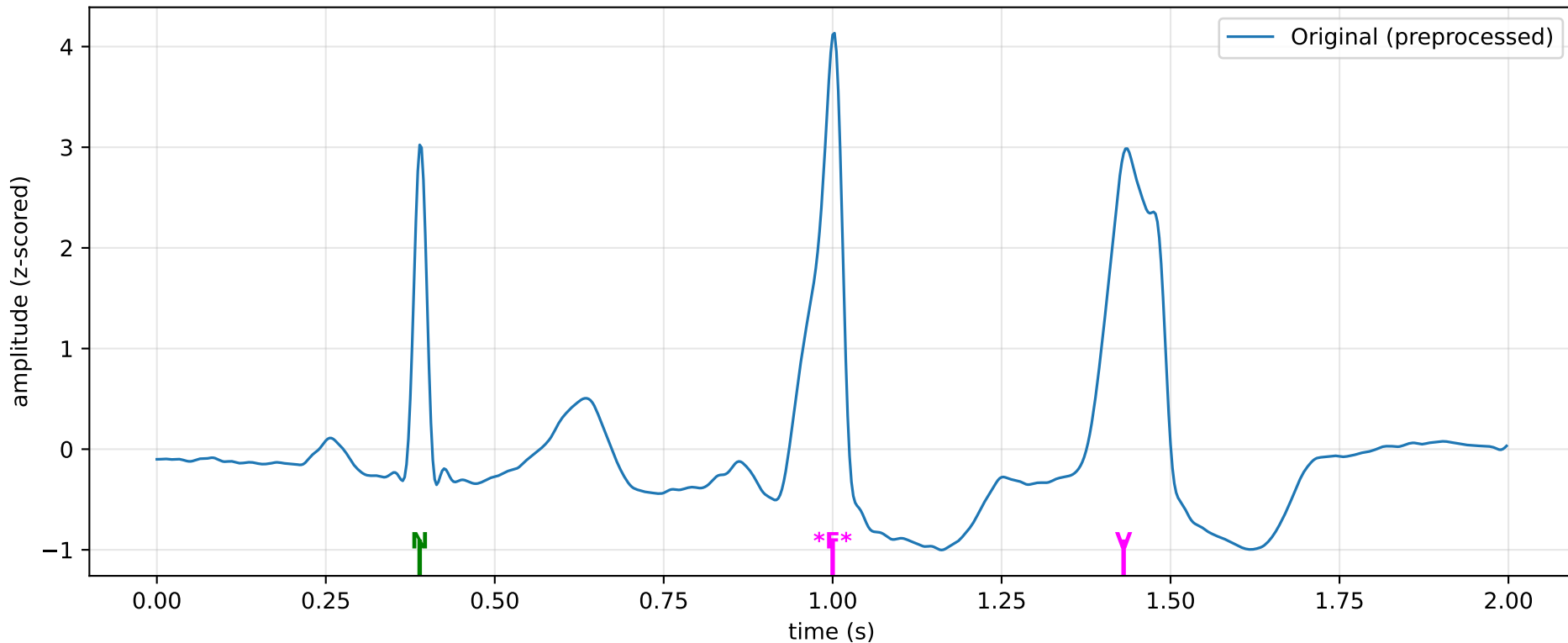
FN: Fusion of ventricular and normal (F)



Symbol	F (centre beat - the one that labels the window)
Type	Fusion of ventricular and normal
Source	record 213, sample 406607
In window	-553ms:N +0ms:*F* +561ms:N
Residual	0.0431 Threshold 0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN mismatch

NORMAL

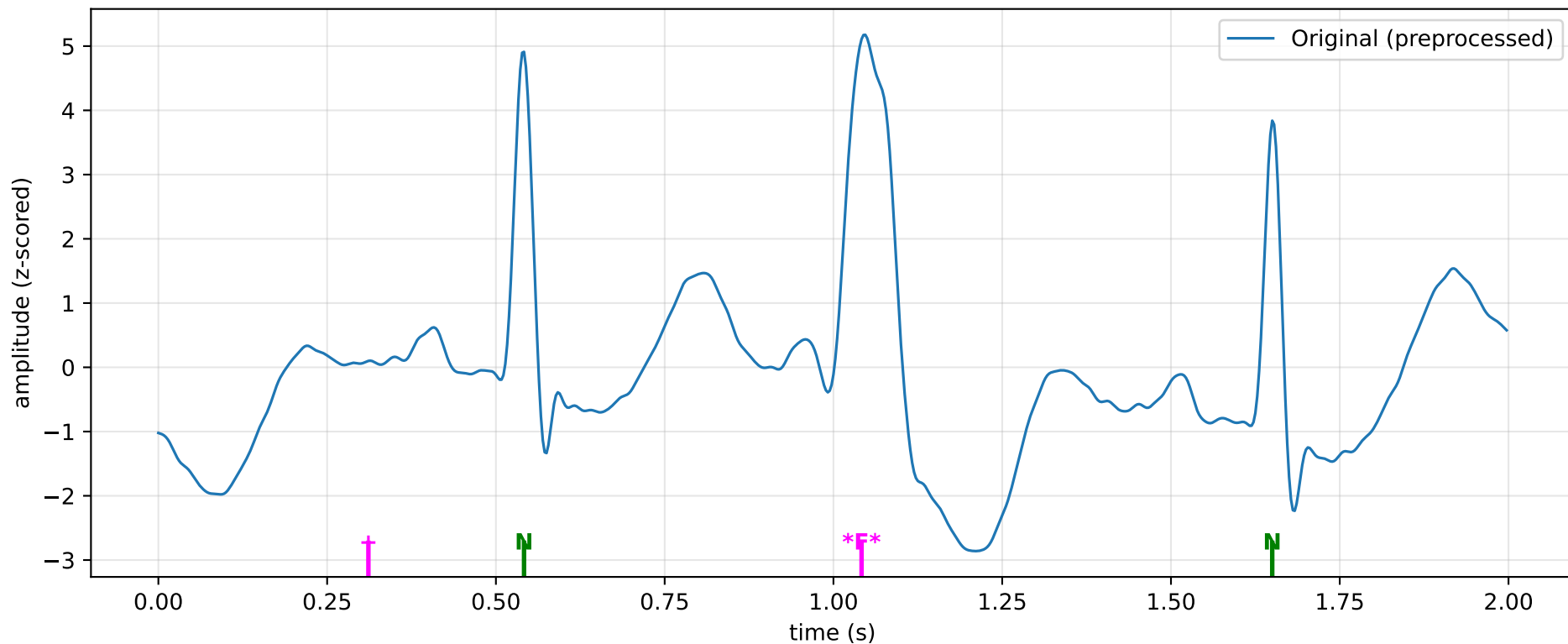
FN: Fusion of ventricular and normal (F)



Symbol	F (centre beat - the one that labels the window)
Type	Fusion of ventricular and normal
Source	record 208, sample 285080
In window	-611ms:N +0ms:*F* +431ms:V
Residual	0.0431 Threshold 0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN mismatch

NORMAL

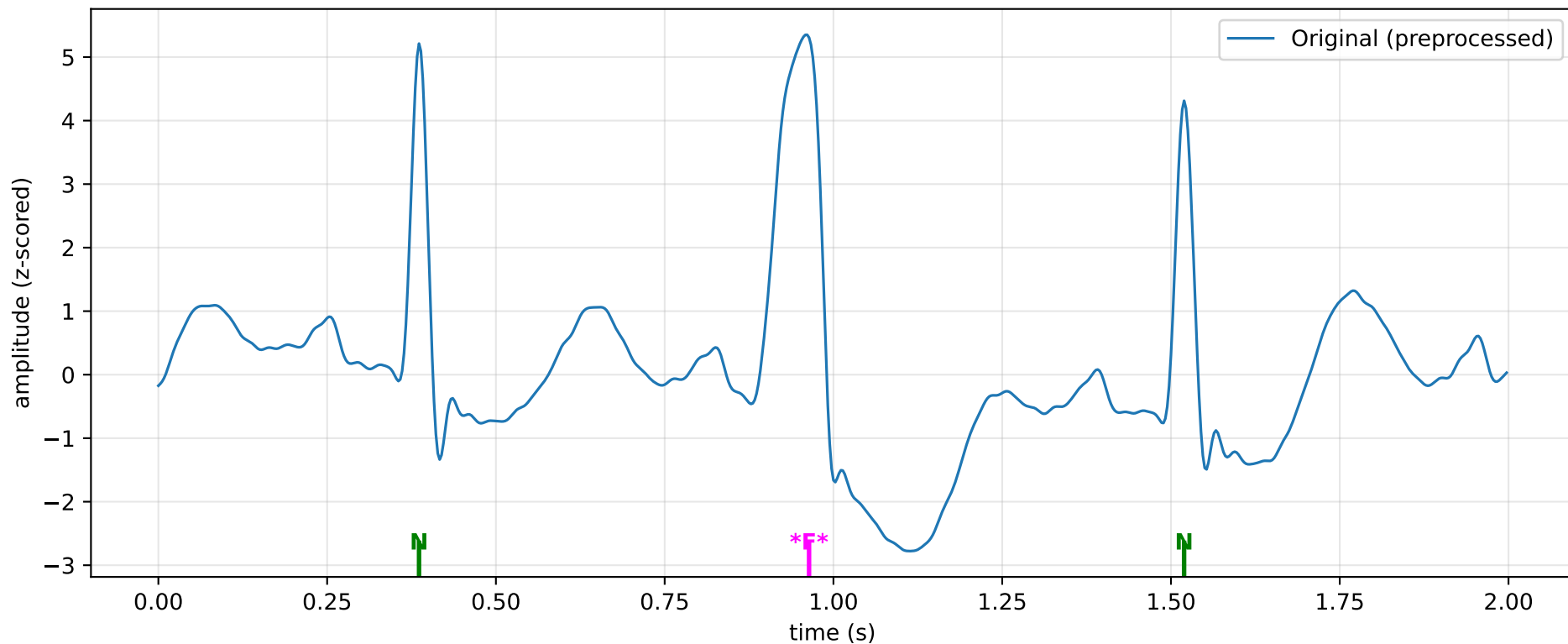
TP: Fusion of ventricular and normal (F)



Symbol	F (centre beat - the one that labels the window)
Type	Fusion of ventricular and normal
Source	record 213, sample 380292
In window	-689ms:+ -458ms:N +42ms:*F* +650ms:N
Residual	0.6744 Threshold 0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP agree

ANOMALY

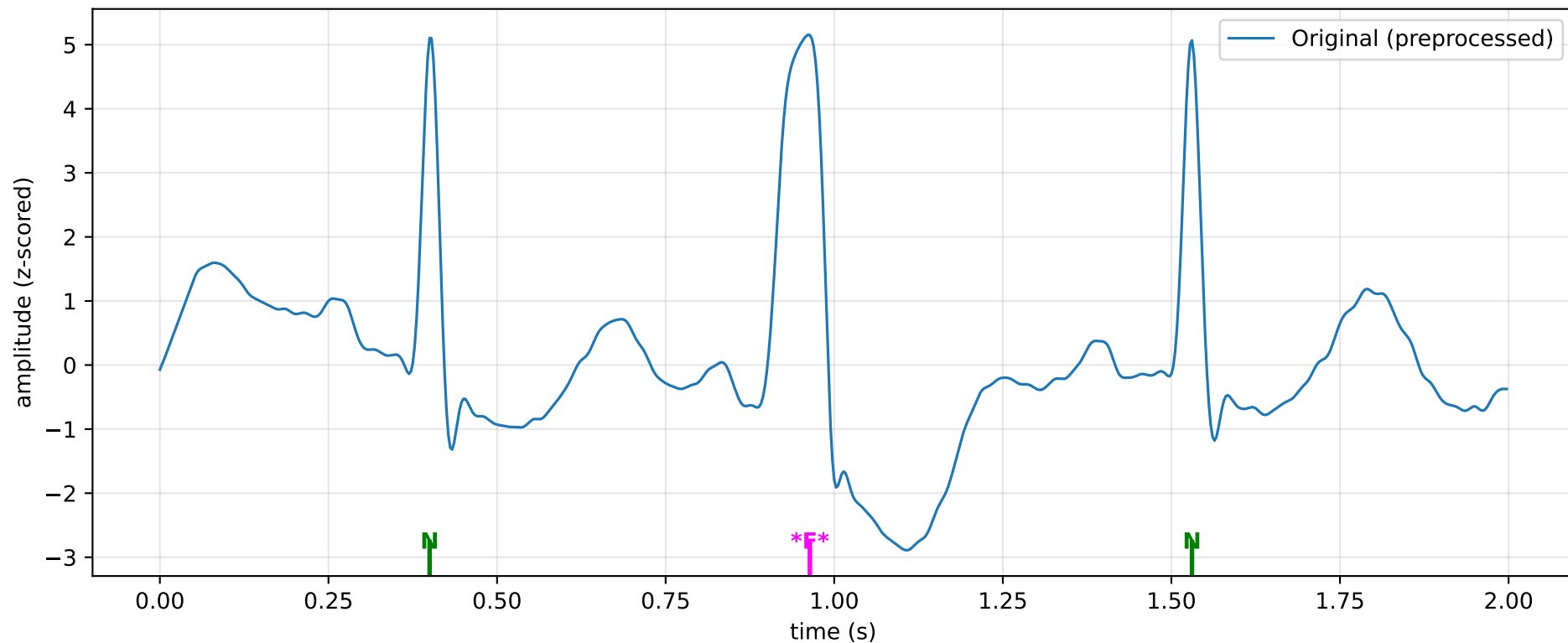
TP: Fusion of ventricular and normal (F)



Symbol	F (centre beat - the one that labels the window)
Type	Fusion of ventricular and normal
Source	record 213, sample 477982
In window	-614ms:N -36ms:*F* +519ms:N
Residual	0.2730 Threshold 0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP agree

ANOMALY

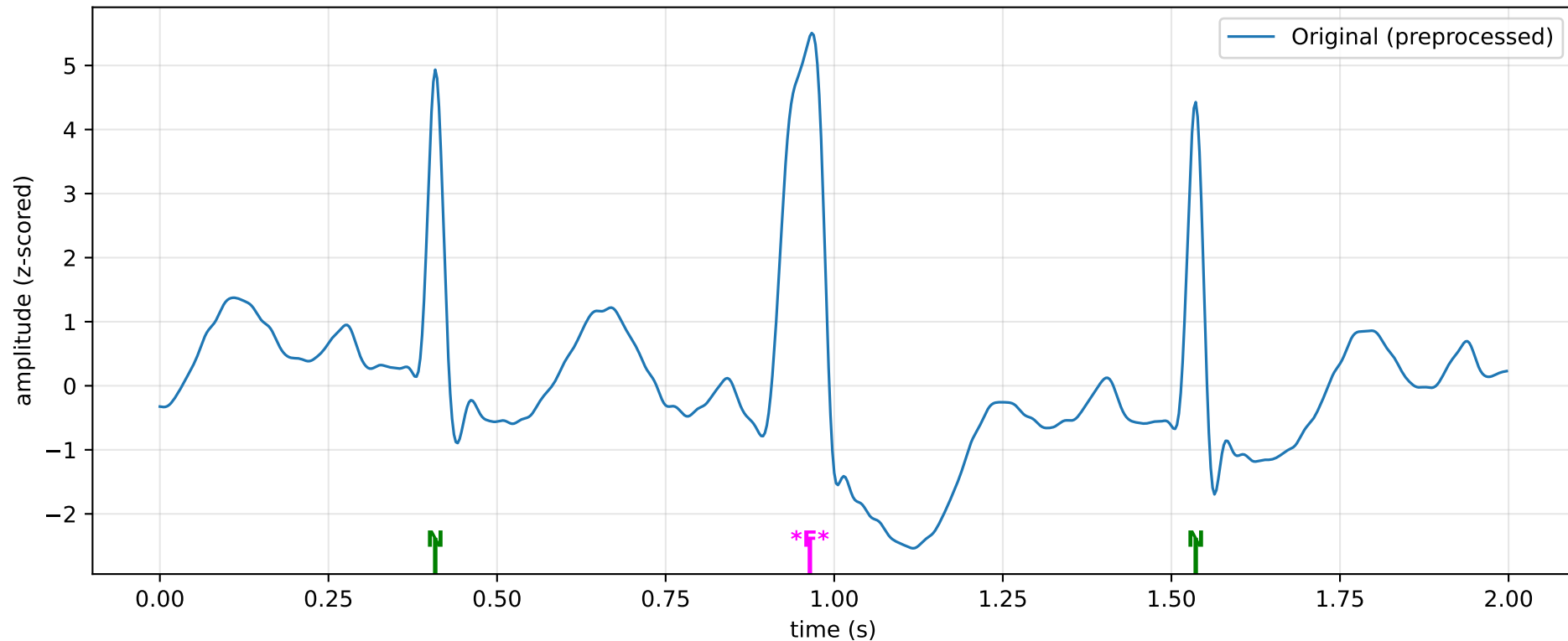
TP: Fusion of ventricular and normal (F)



Symbol	F (centre beat - the one that labels the window)
Type	Fusion of ventricular and normal
Source	record 213, sample 483675
In window	-600ms:N -36ms:*F* +531ms:N
Residual	0.2259 Threshold 0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP agree

ANOMALY

TP: Fusion of ventricular and normal (F)

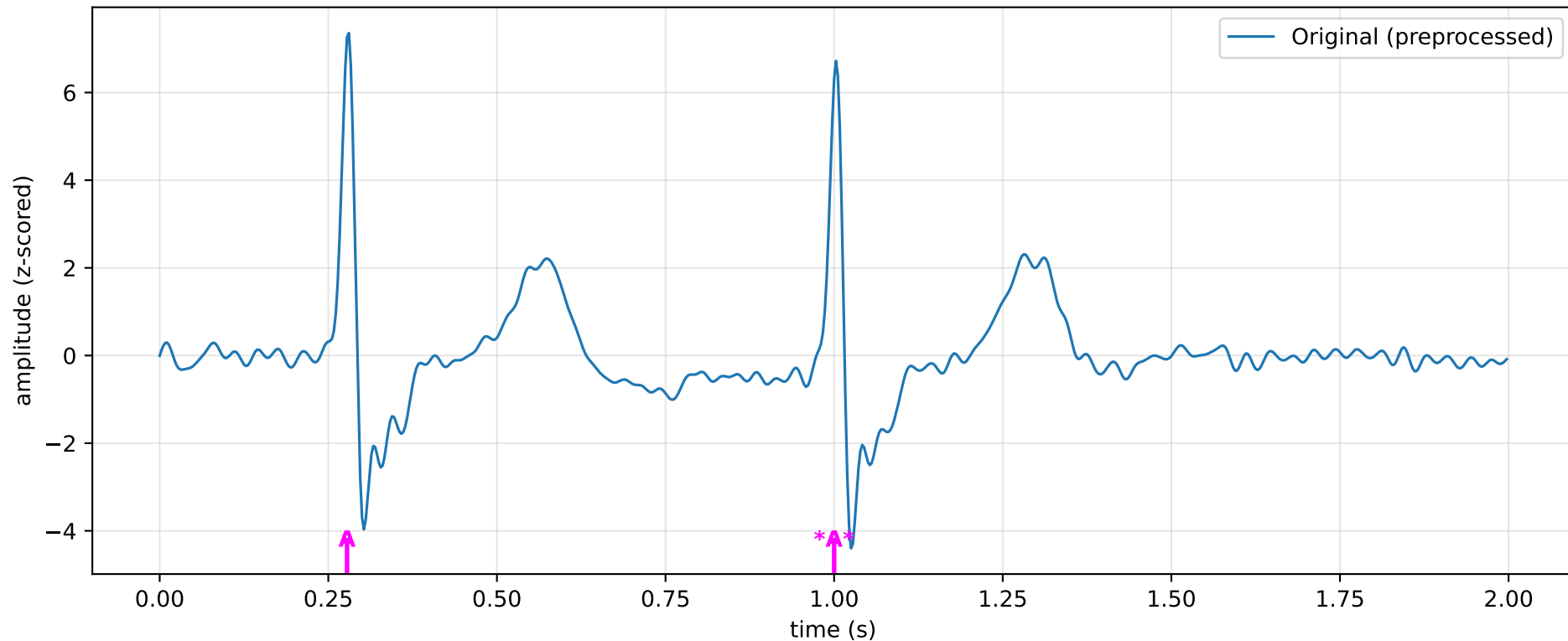


Symbol	F (centre beat - the one that labels the window)
Type	Fusion of ventricular and normal
Source	record 213, sample 485711
In window	-592ms:N -36ms:*F* +536ms:N
Residual	0.2060 Threshold 0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP agree

ANOMALY

Section: A - Atrial premature beat (APC)

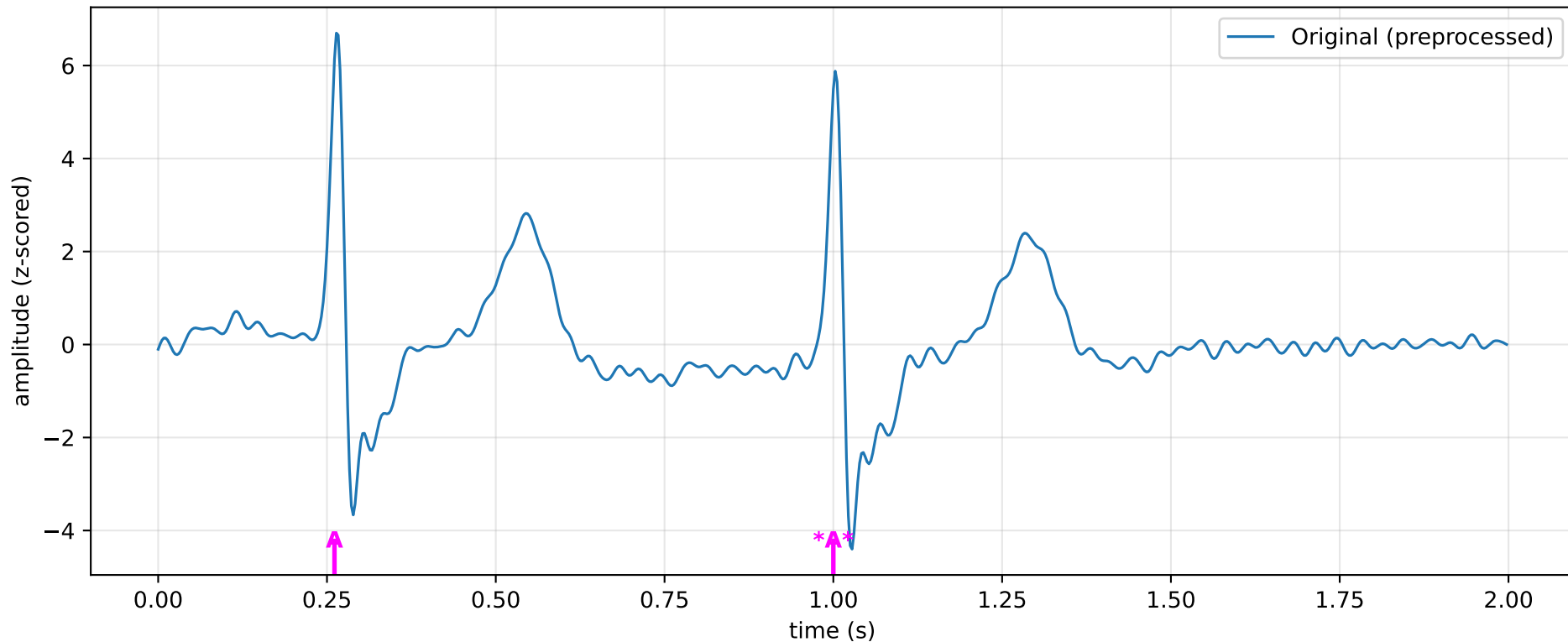
FN: Atrial premature beat (APC) (A)



Symbol	A (centre beat - the one that labels the window)
Type	Atrial premature beat (APC)
Source	record 232, sample 375665
In window	-722ms:A +0ms:*A*
Residual	0.0434 Threshold 0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN mismatch

NORMAL

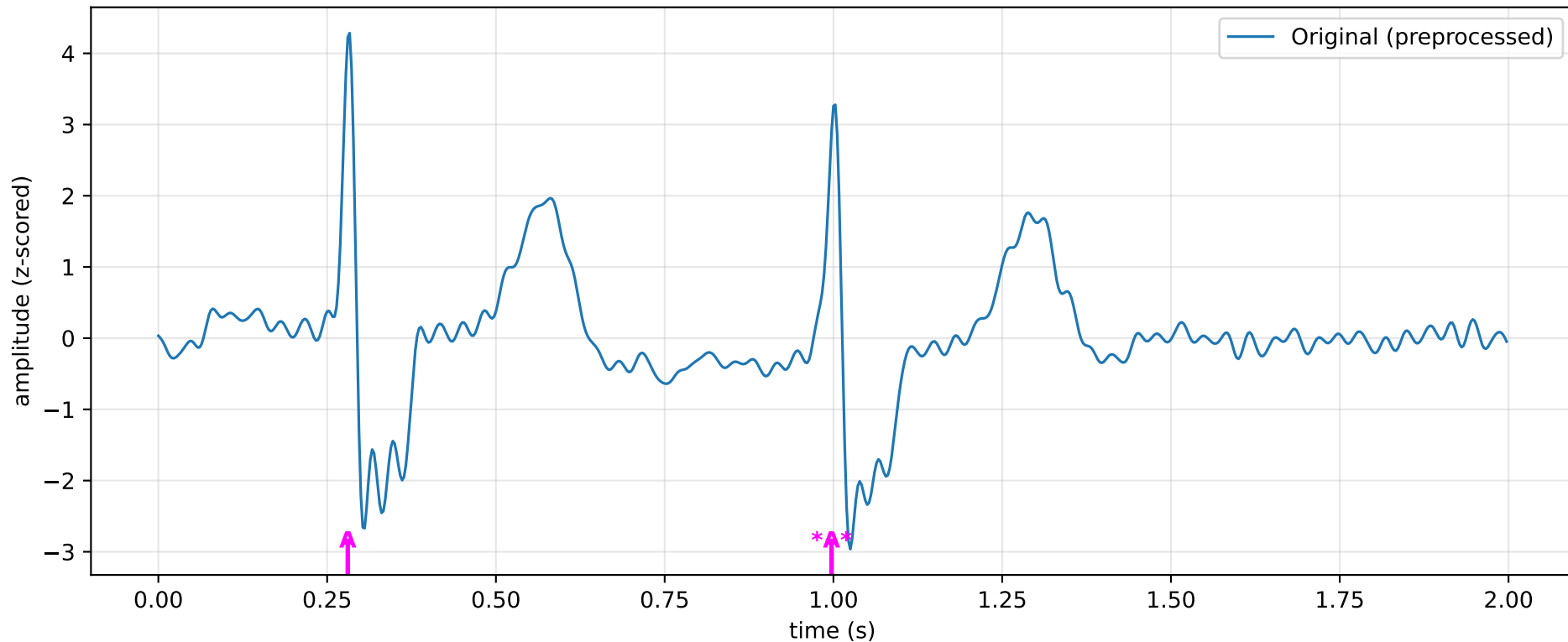
FN: Atrial premature beat (APC) (A)



Symbol	A (centre beat - the one that labels the window)
Type	Atrial premature beat (APC)
Source	record 232, sample 285562
In window	-739ms:A +0ms:*A*
Residual	0.0433 Threshold 0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN mismatch

NORMAL

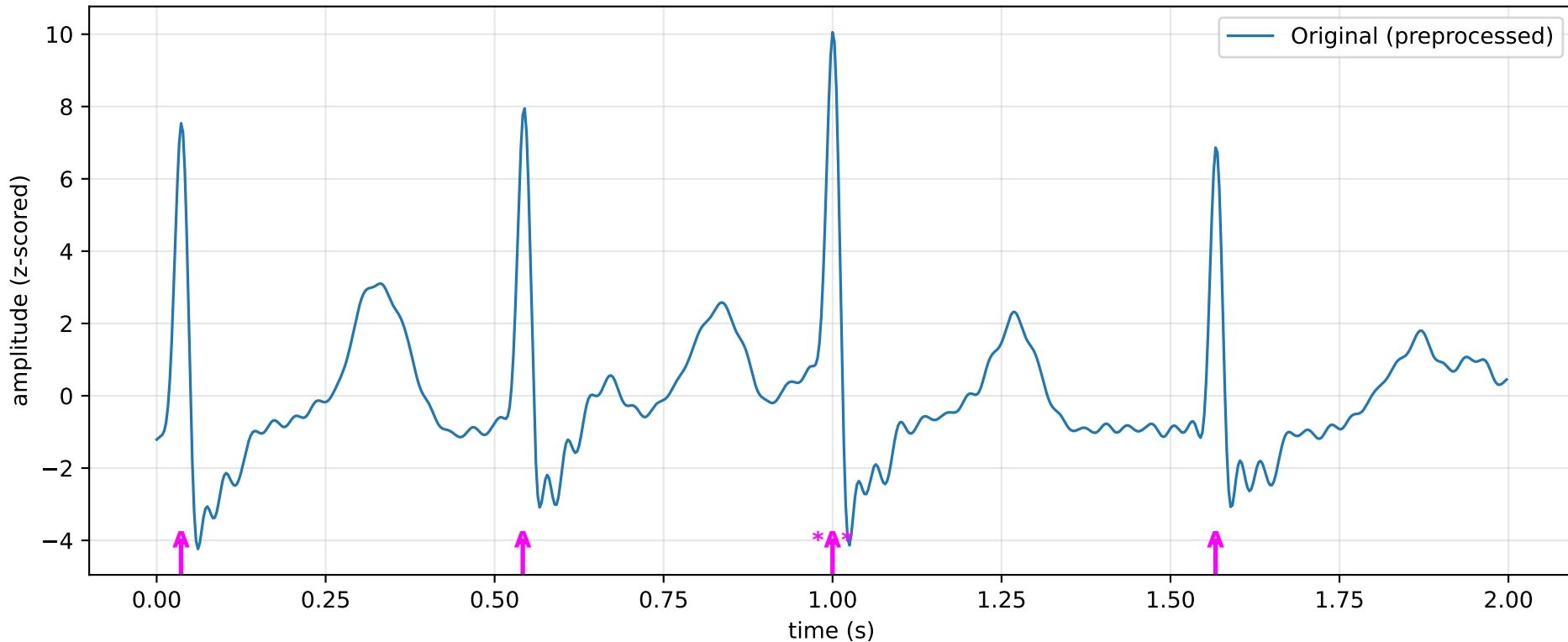
FN: Atrial premature beat (APC) (A)



Symbol	A (centre beat - the one that labels the window)
Type	Atrial premature beat (APC)
Source	record 232, sample 40168
In window	-719ms:A -3ms:*A*
Residual	0.0433 Threshold 0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN mismatch

NORMAL

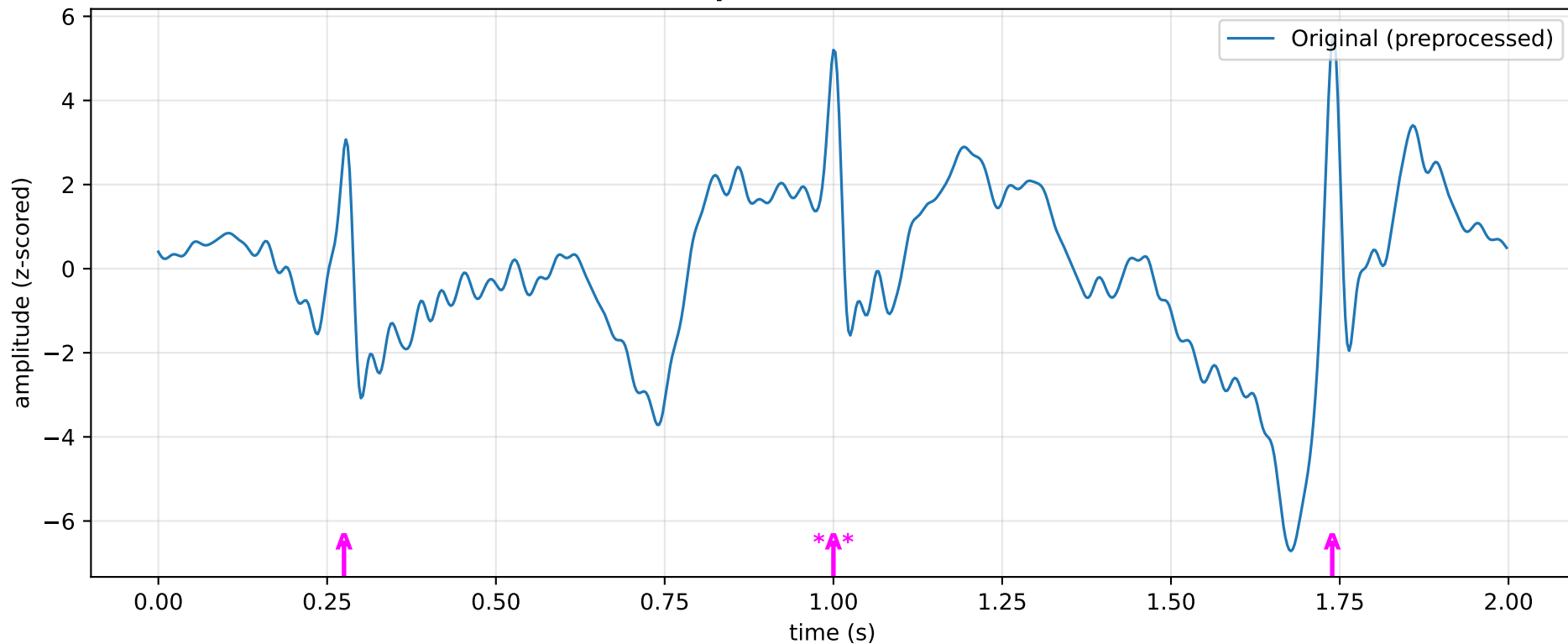
TP: Atrial premature beat (APC) (A)



Symbol	A (centre beat - the one that labels the window)			
Type	Atrial premature beat (APC)			
Source	record 232, sample 446678			
In window	-964ms:A	-458ms:A	+0ms:*A*	+567ms:A
Residual	1.2566	Threshold 0.0434		
Decision	ANOMALY (truth: ANOMALY)			
Outcome	TP agree			

ANOMALY

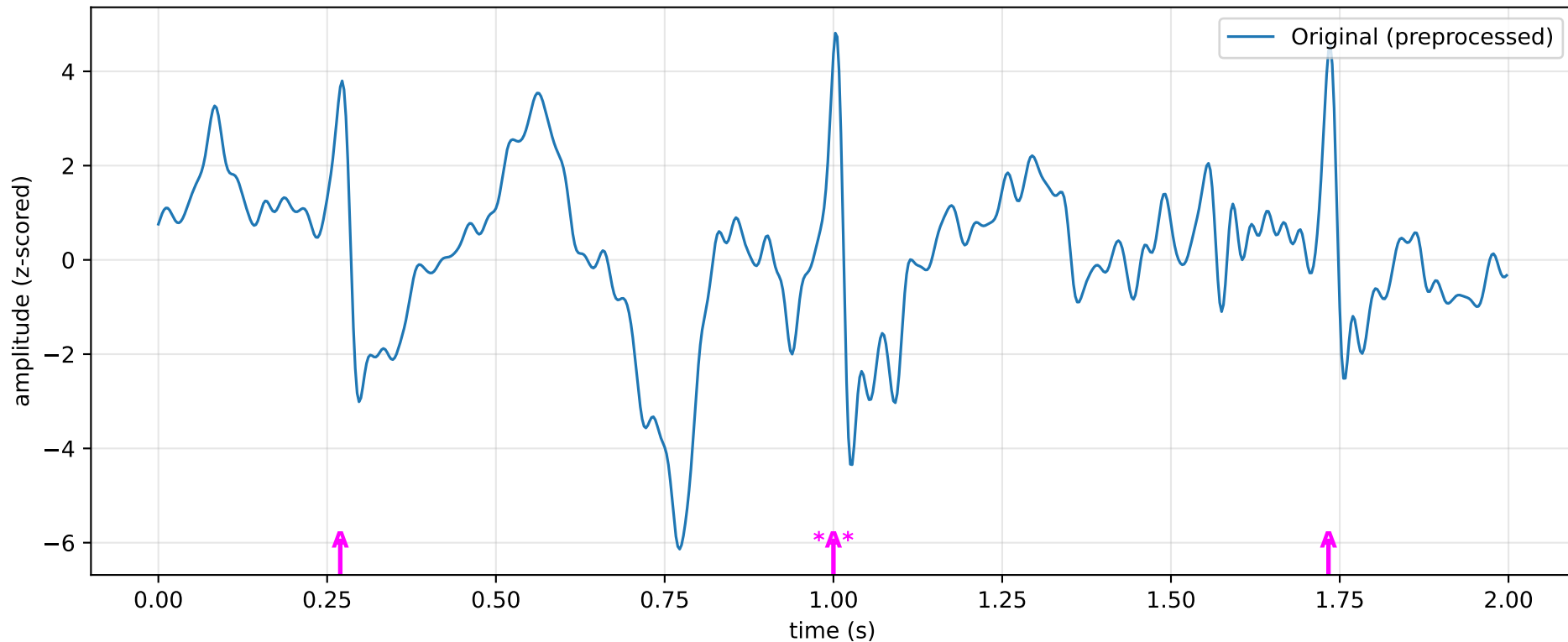
TP: Atrial premature beat (APC) (A)



Symbol	A (centre beat - the one that labels the window)
Type	Atrial premature beat (APC)
Source	record 232, sample 245833
In window	-725ms:A +0ms:*A* +739ms:A
Residual	1.2324 Threshold 0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP agree

ANOMALY

TP: Atrial premature beat (APC) (A)

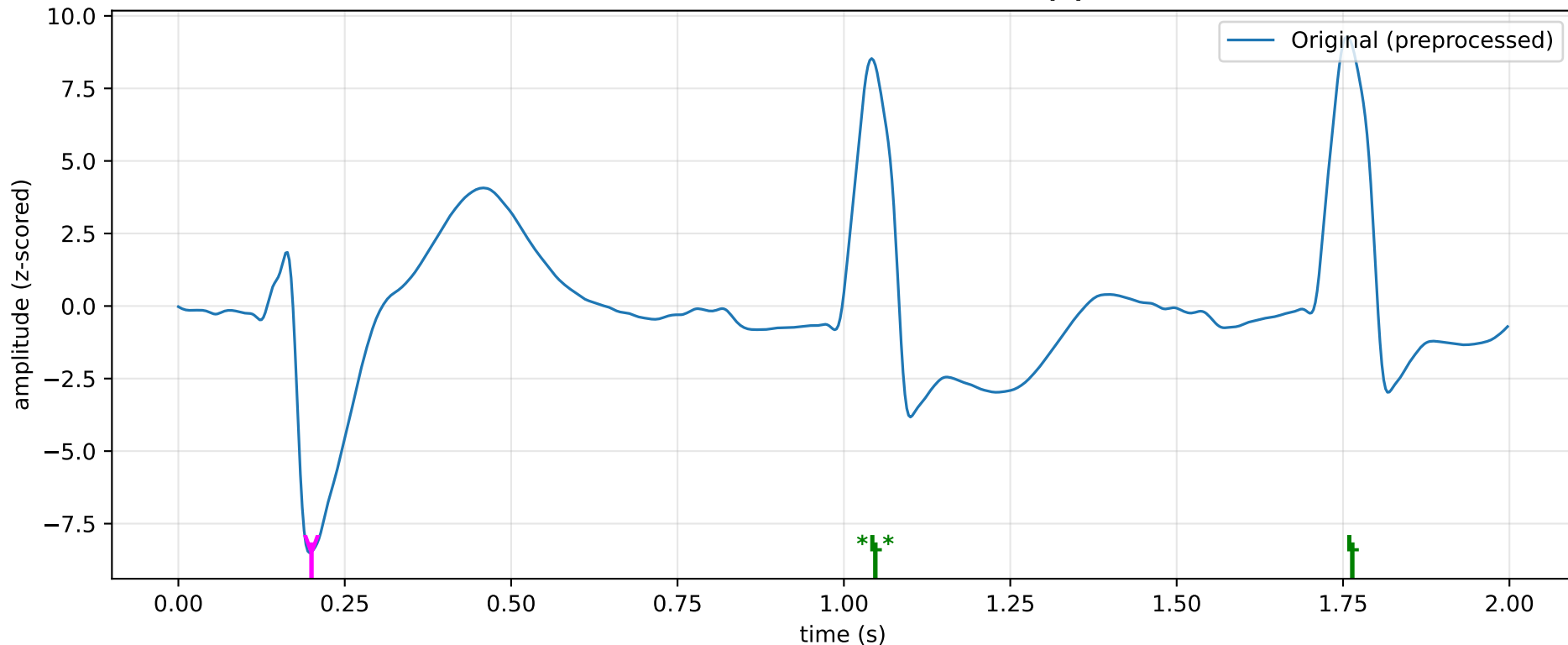


Symbol	A (centre beat - the one that labels the window)
Type	Atrial premature beat (APC)
Source	record 232, sample 247683
In window	-731ms:A +0ms:*A* +733ms:A
Residual	0.9413 Threshold 0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP agree

ANOMALY

Section: L - Left bundle branch block

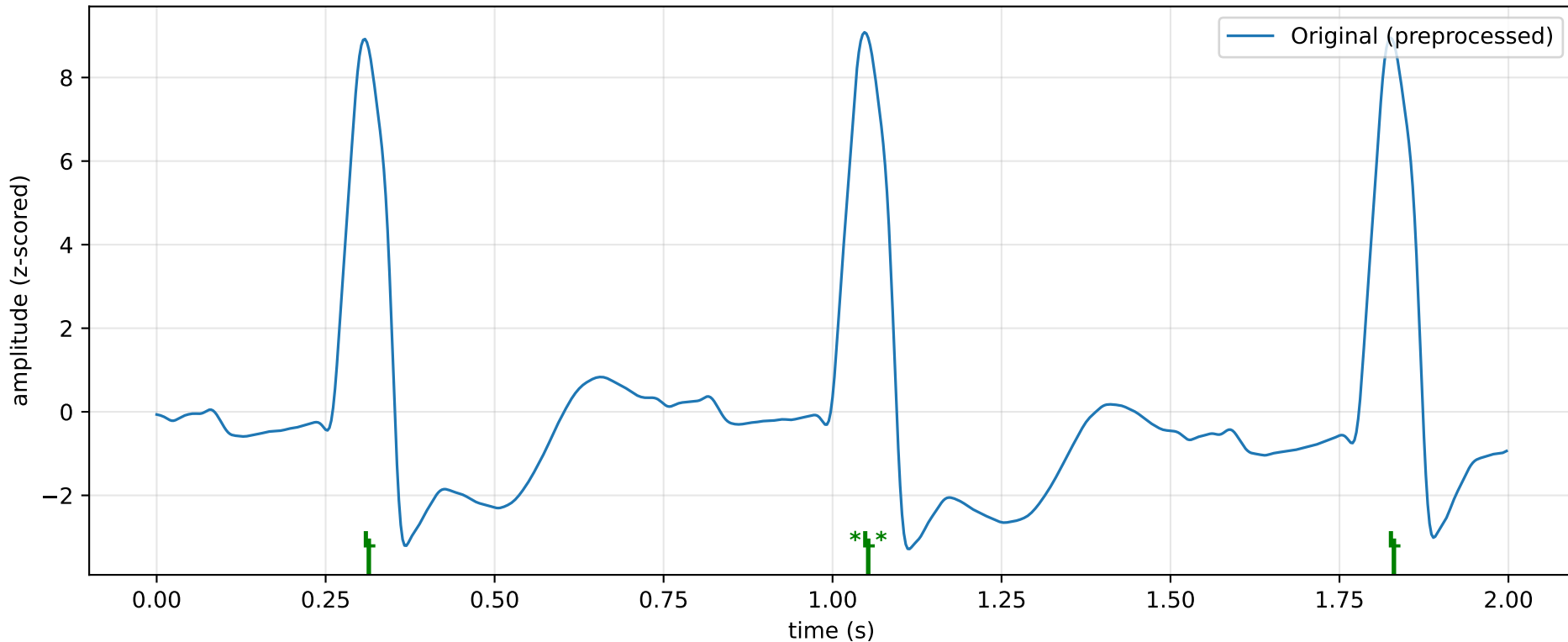
FP: Left bundle branch block (L)



Symbol	L (centre beat - the one that labels the window)
Type	Left bundle branch block
Source	record 109, sample 583956
In window	-800ms:V +47ms:*L* +764ms:L
Residual	1.7865 Threshold 0.0434
Decision	ANOMALY (truth: NORMAL)
Outcome	FP mismatch

ANOMALY

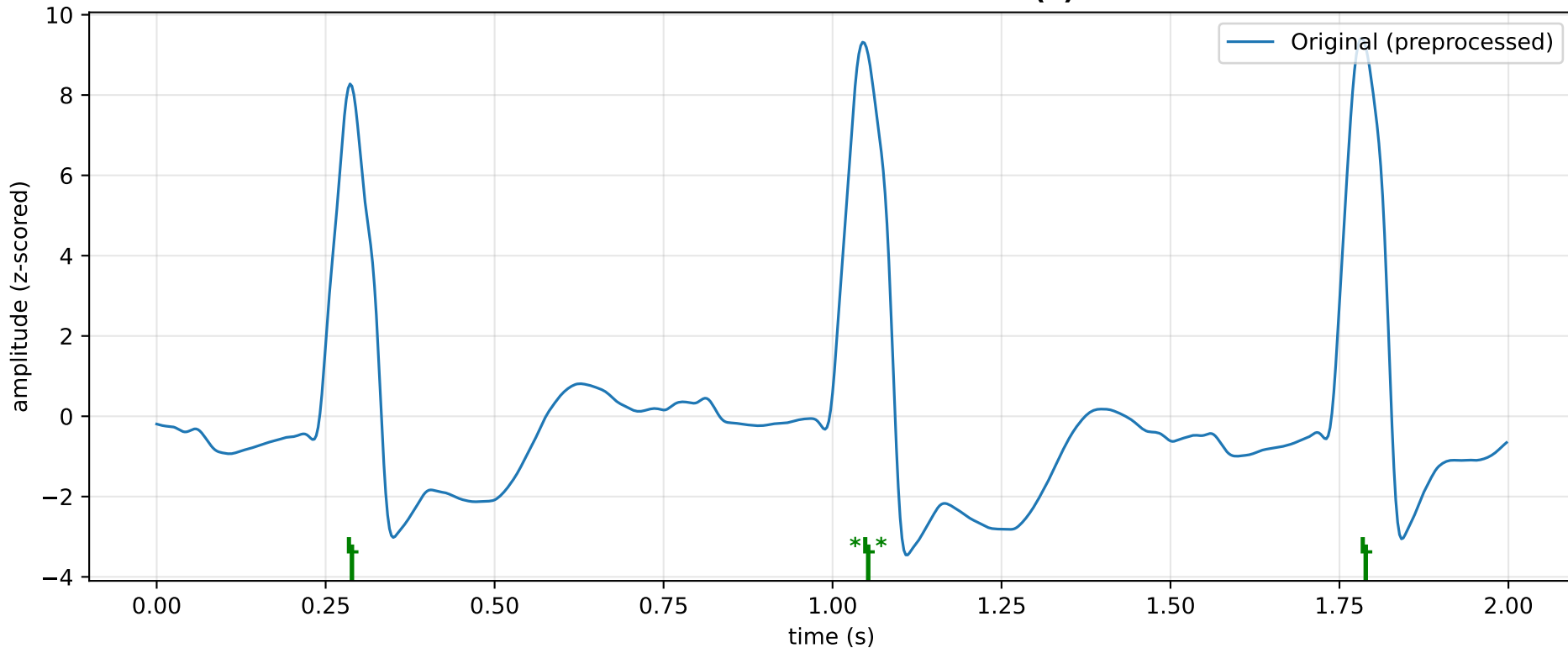
FP: Left bundle branch block (L)



Symbol	L (centre beat - the one that labels the window)
Type	Left bundle branch block
Source	record 109, sample 625192
In window	-686ms:L +53ms:*L* +831ms:L
Residual	1.6827 Threshold 0.0434
Decision	ANOMALY (truth: NORMAL)
Outcome	FP mismatch

ANOMALY

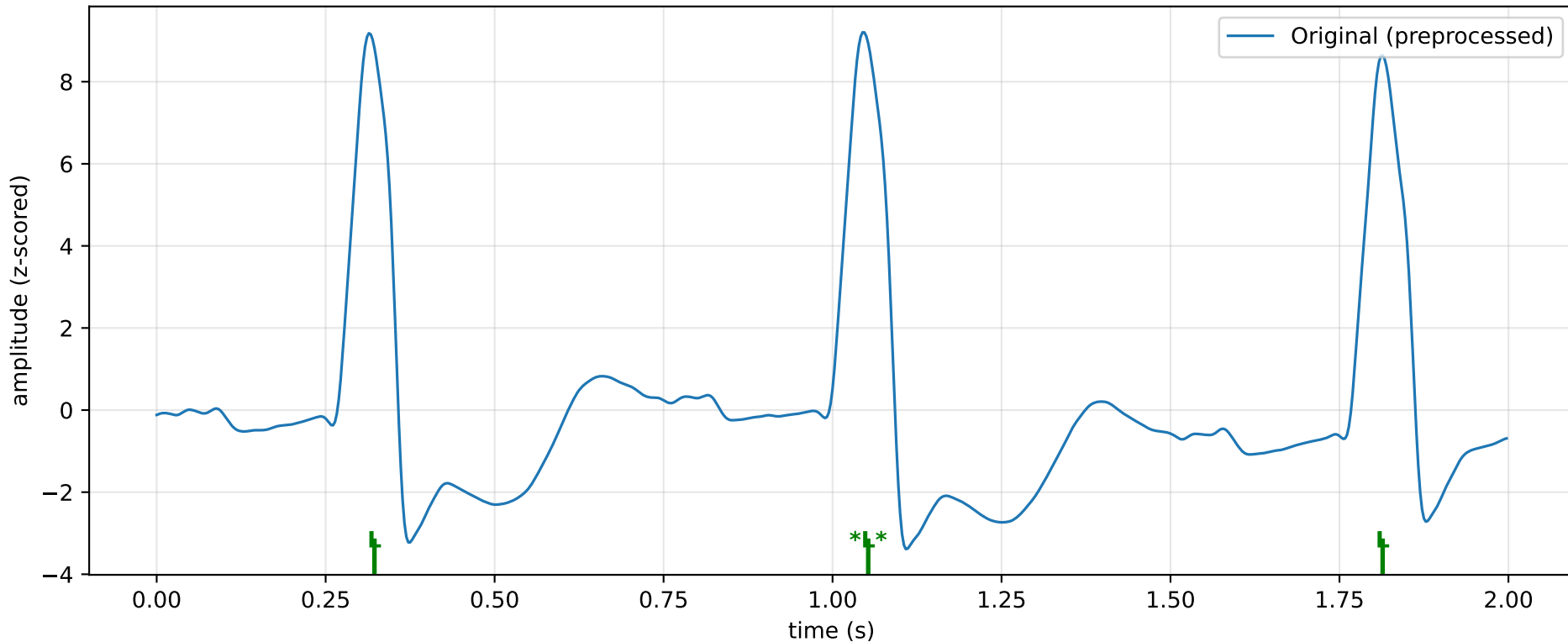
FP: Left bundle branch block (L)



Symbol	L (centre beat - the one that labels the window)
Type	Left bundle branch block
Source	record 109, sample 604399
In window	-711ms:L +53ms:*L* +789ms:L
Residual	1.6012 Threshold 0.0434
Decision	ANOMALY (truth: NORMAL)
Outcome	FP mismatch

ANOMALY

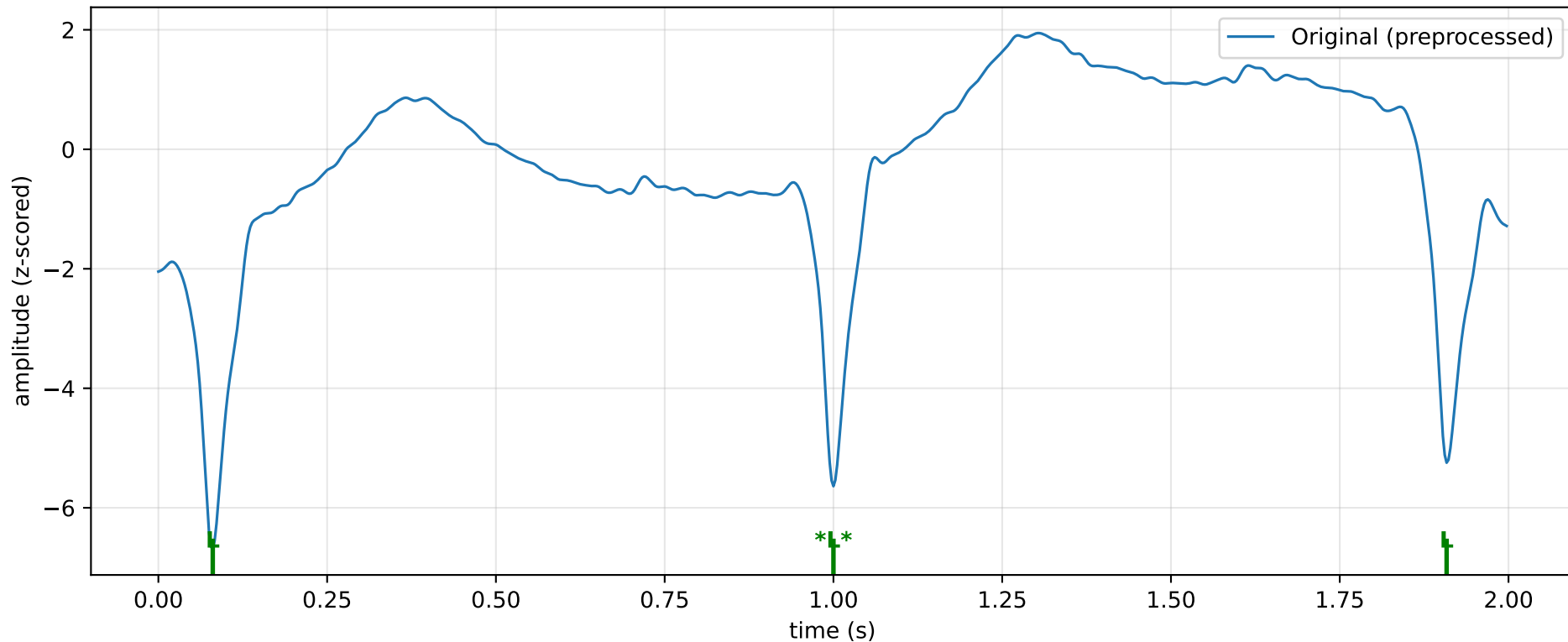
FP: Left bundle branch block (L)



Symbol	L (centre beat - the one that labels the window)
Type	Left bundle branch block
Source	record 109, sample 601472
In window	-678ms:L +53ms:*L* +814ms:L
Residual	1.5828 Threshold 0.0434
Decision	ANOMALY (truth: NORMAL)
Outcome	FP mismatch

ANOMALY

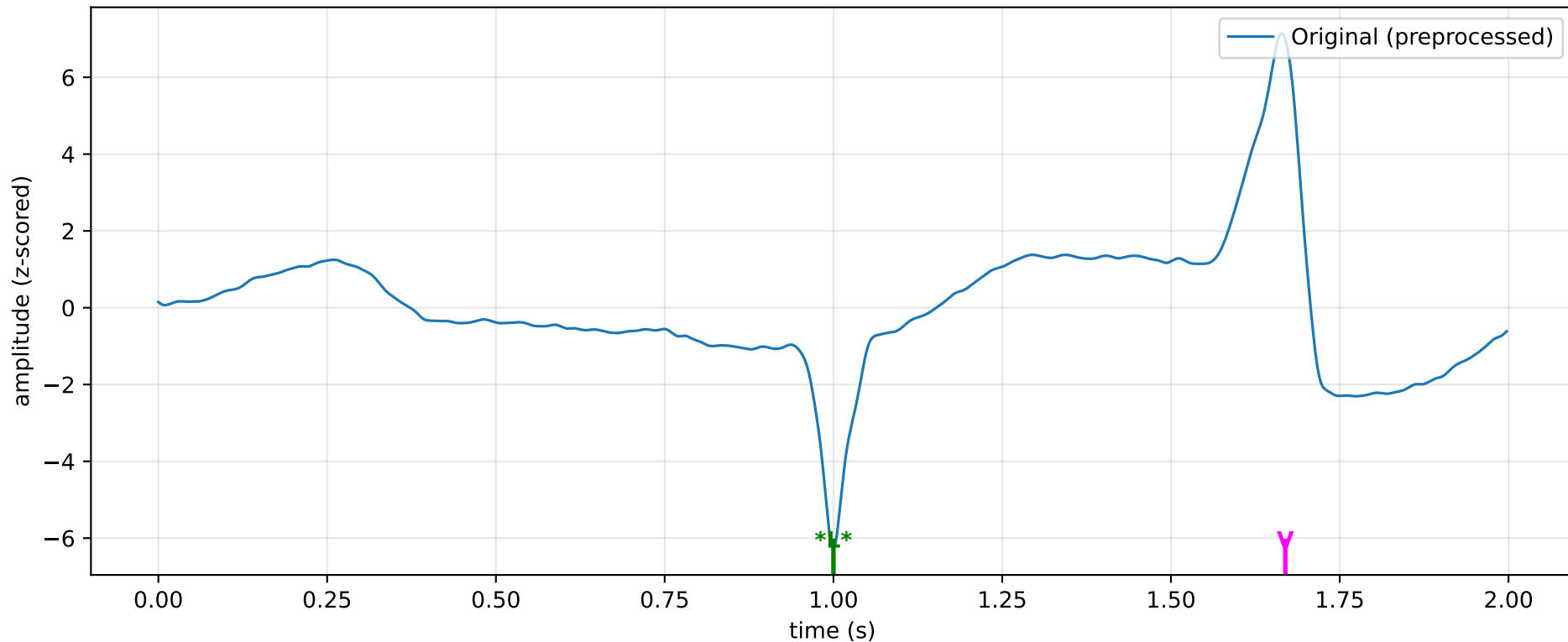
TN: Left bundle branch block (L)



Symbol	L (centre beat - the one that labels the window)
Type	Left bundle branch block
Source	record 207, sample 288750
In window	-919ms:L +0ms:*L* +908ms:L
Residual	0.0433 Threshold 0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN agree

NORMAL

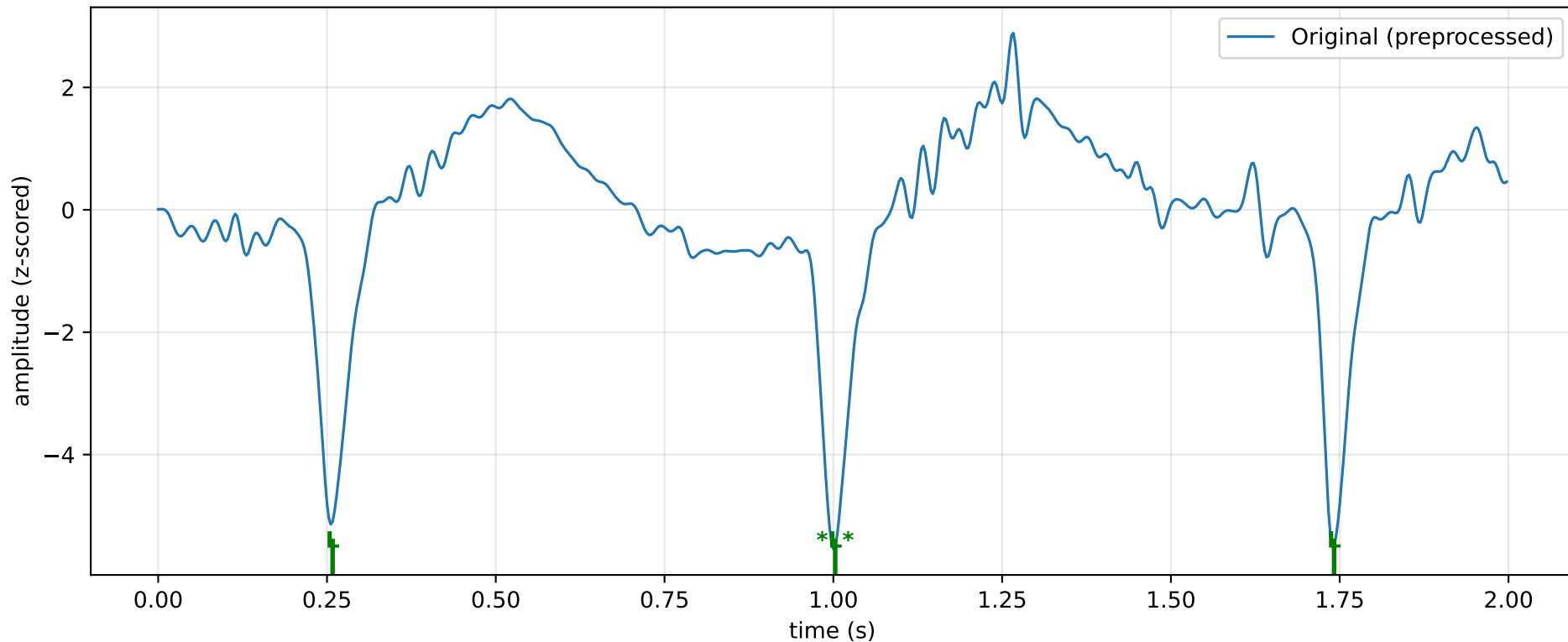
TN: Left bundle branch block (L)



Symbol	L (centre beat - the one that labels the window)
Type	Left bundle branch block
Source	record 207, sample 29847
In window	+0ms:*L* +669ms:V
Residual	0.0433 Threshold 0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN agree

NORMAL

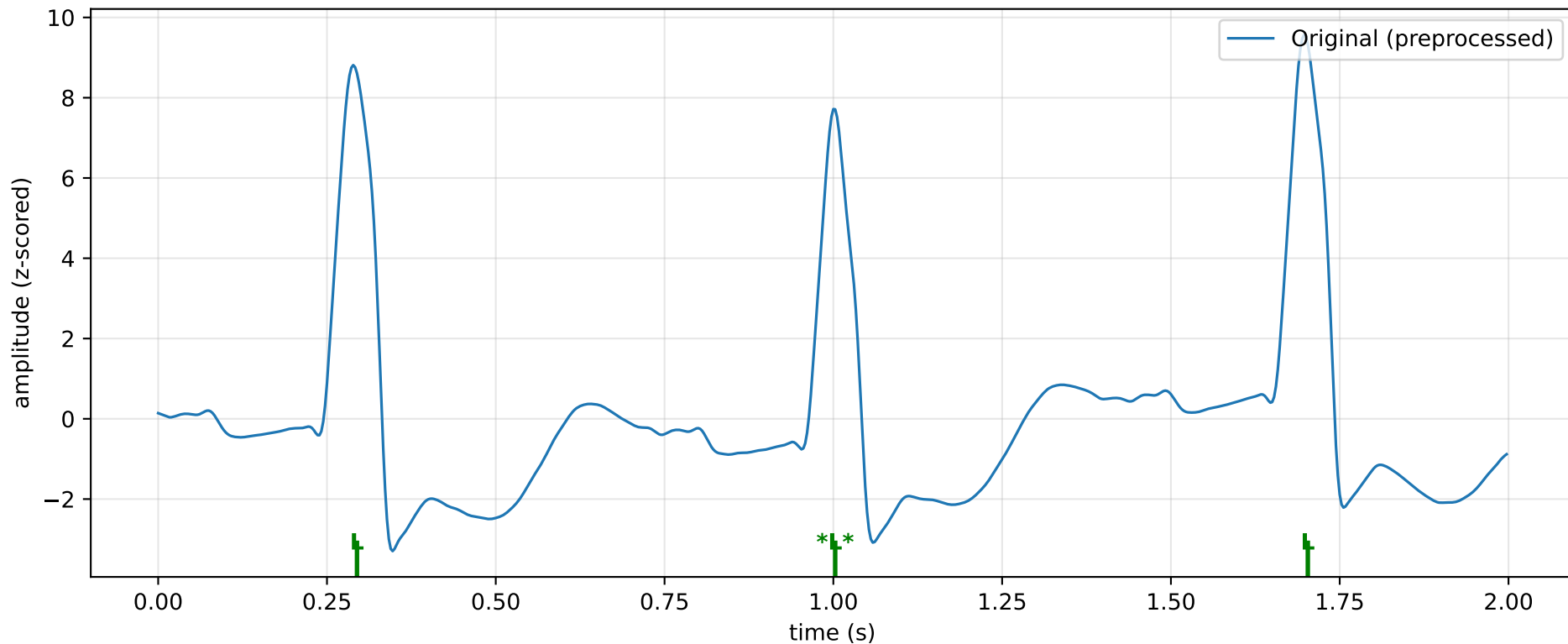
TN: Left bundle branch block (L)



Symbol	L (centre beat - the one that labels the window)
Type	Left bundle branch block
Source	record 207, sample 113694
In window	-742ms:L +3ms:*L* +742ms:L
Residual	0.0433 Threshold 0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN agree

NORMAL

TN: Left bundle branch block (L)

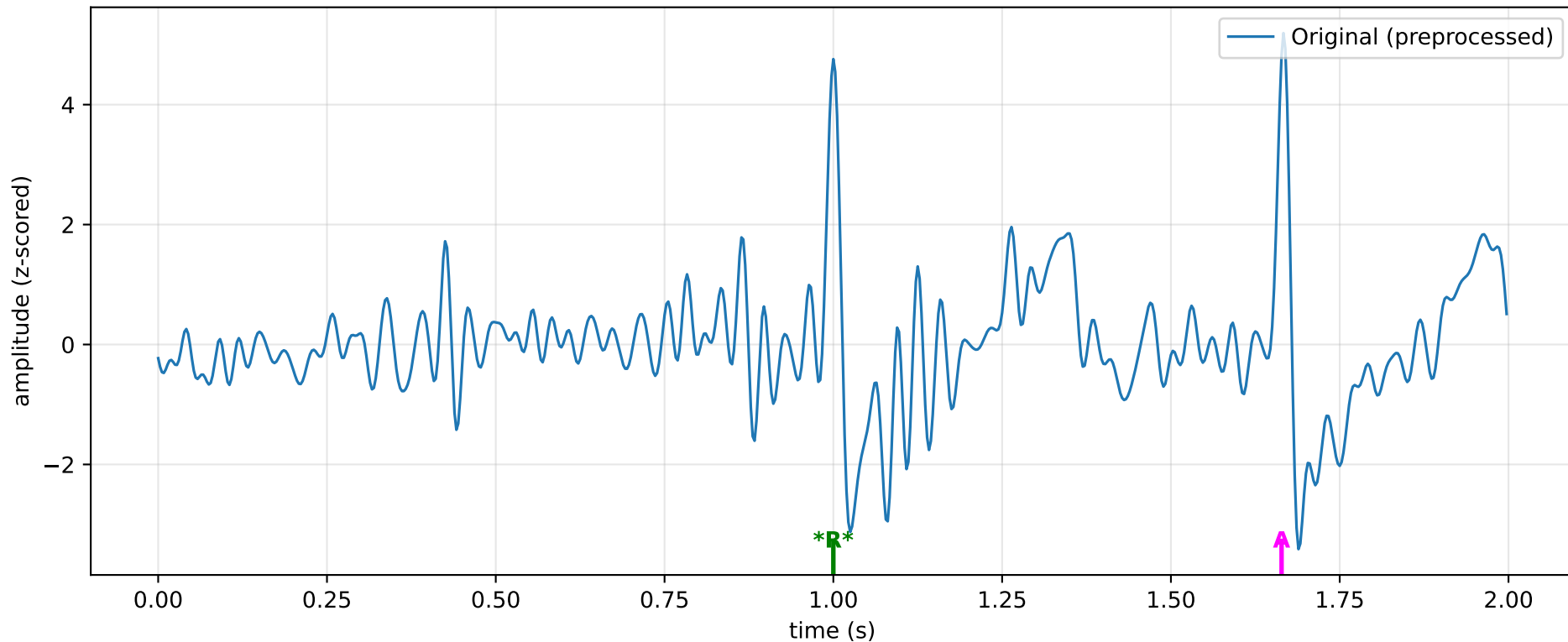


Symbol	L (centre beat - the one that labels the window)
Type	Left bundle branch block
Source	record 109, sample 619904
In window	-706ms:L +3ms:*L* +703ms:L
Residual	0.0433 Threshold 0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN agree

NORMAL

Section: R - Right bundle branch block

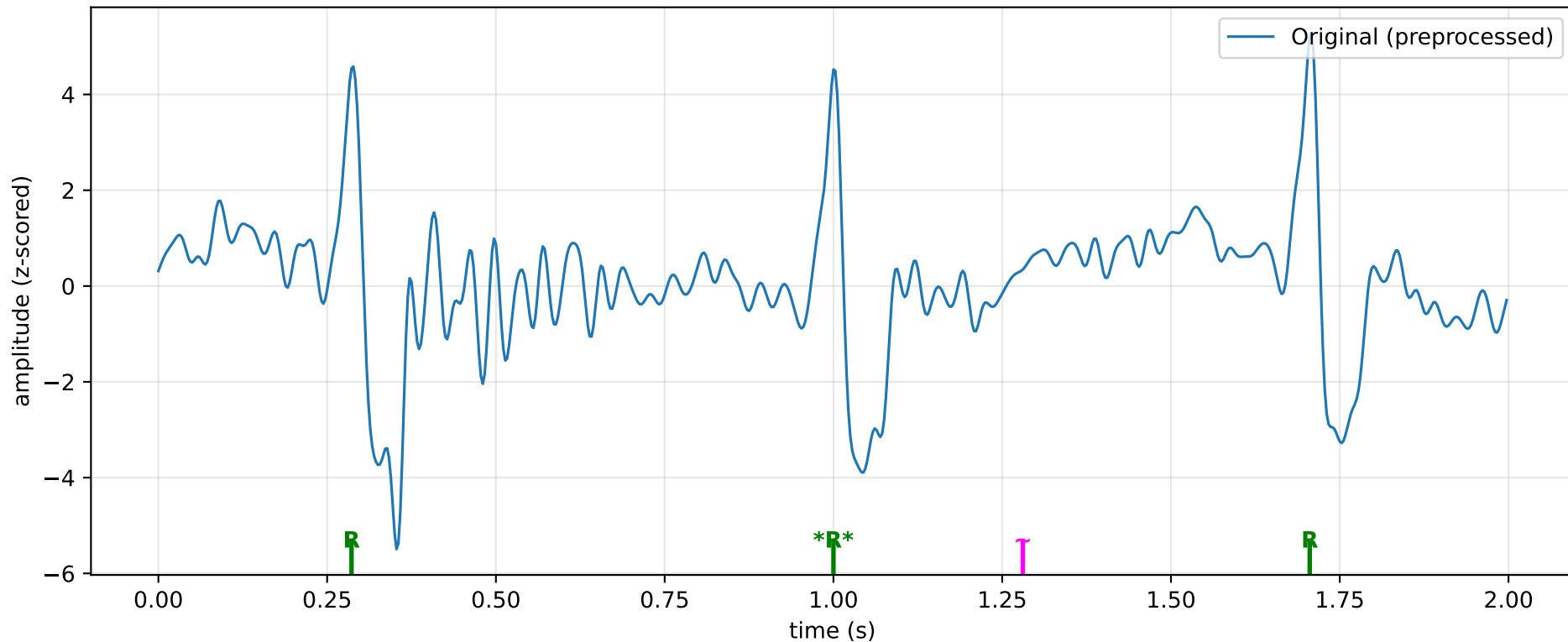
FP: Right bundle branch block (R)



Symbol	R (centre beat - the one that labels the window)
Type	Right bundle branch block
Source	record 232, sample 22726
In window	+0ms:*R* +664ms:A
Residual	0.2474 Threshold 0.0434
Decision	ANOMALY (truth: NORMAL)
Outcome	FP mismatch

ANOMALY

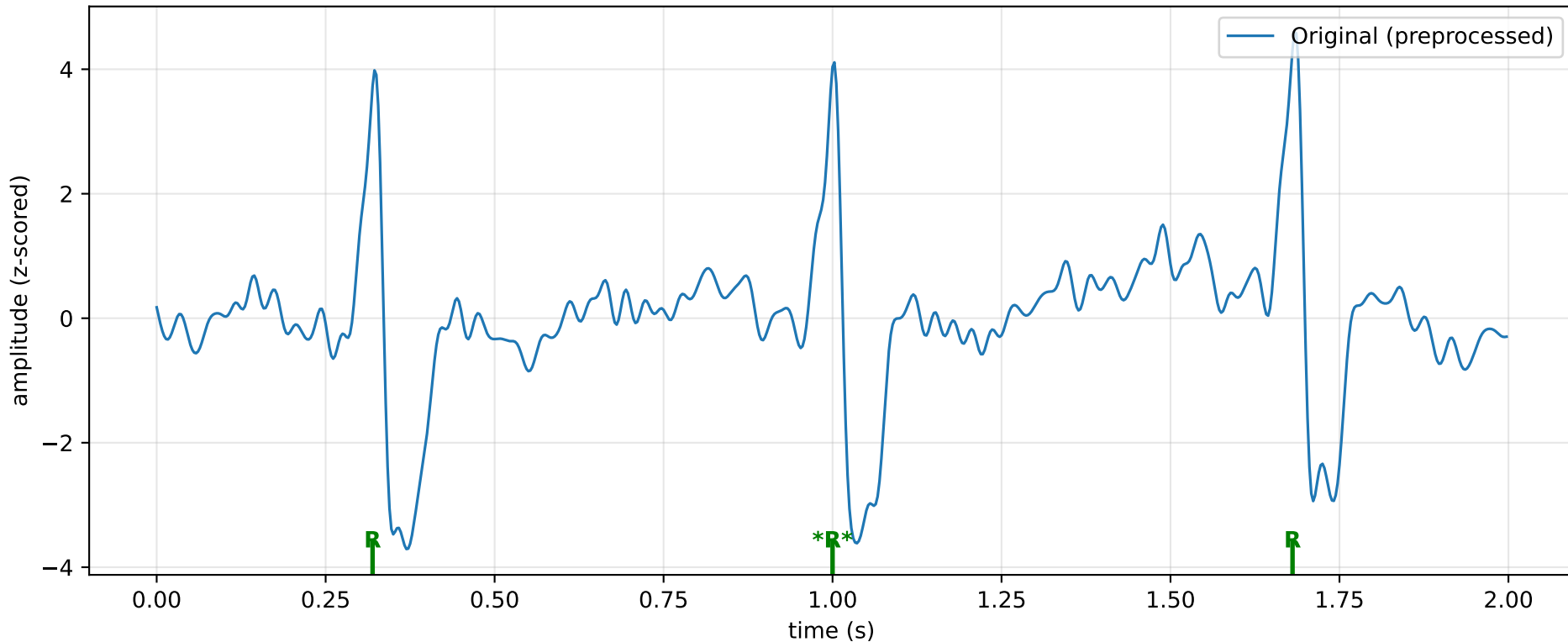
FP: Right bundle branch block (R)



Symbol	R (centre beat - the one that labels the window)
Type	Right bundle branch block
Source	record 118, sample 621965
In window	-714ms:R +0ms:*R* +281ms:~ +706ms:R
Residual	0.2204 Threshold 0.0434
Decision	ANOMALY (truth: NORMAL)
Outcome	FP mismatch

ANOMALY

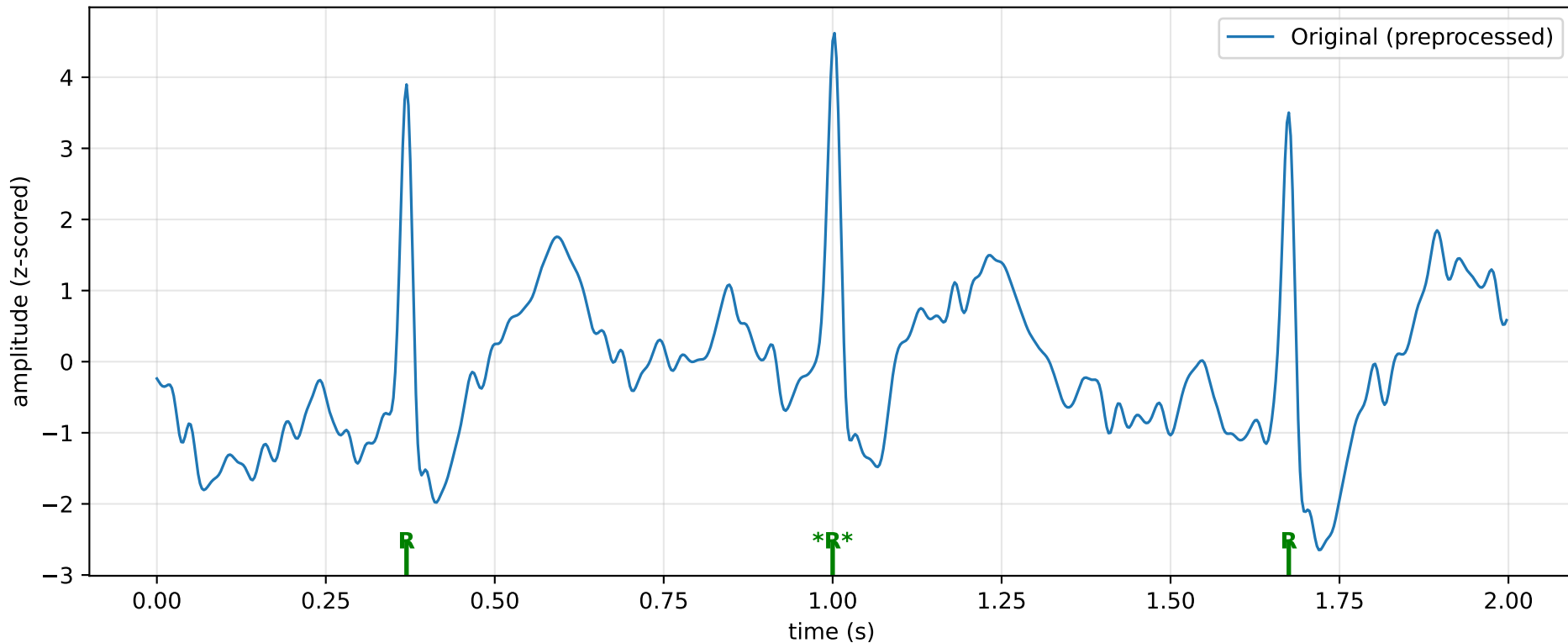
TN: Right bundle branch block (R)



Symbol	R (centre beat - the one that labels the window)
Type	Right bundle branch block
Source	record 118, sample 592155
In window	-681ms:R +0ms:*R* +681ms:R
Residual	0.0434 Threshold 0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN agree

NORMAL

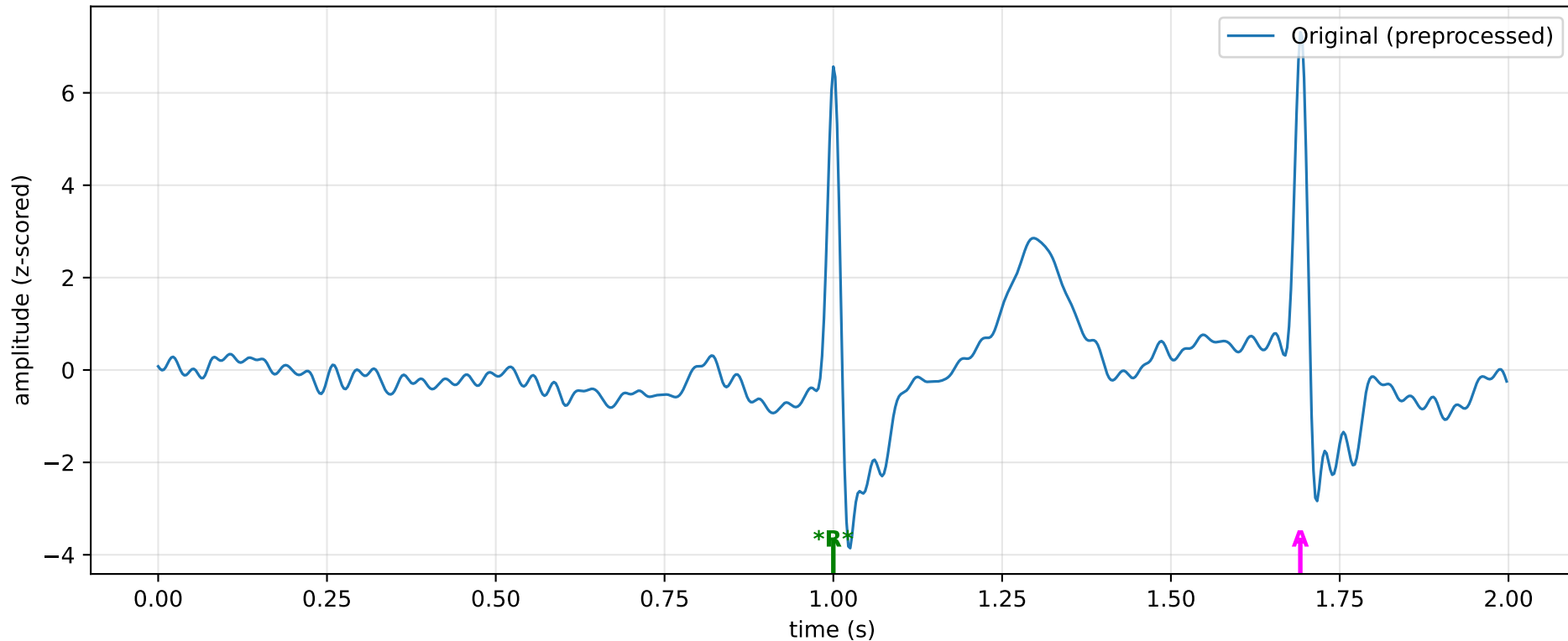
TN: Right bundle branch block (R)



Symbol	R (centre beat - the one that labels the window)
Type	Right bundle branch block
Source	record 212, sample 221336
In window	-631ms:R +0ms:*R* +675ms:R
Residual	0.0433 Threshold 0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN agree

NORMAL

TN: Right bundle branch block (R)

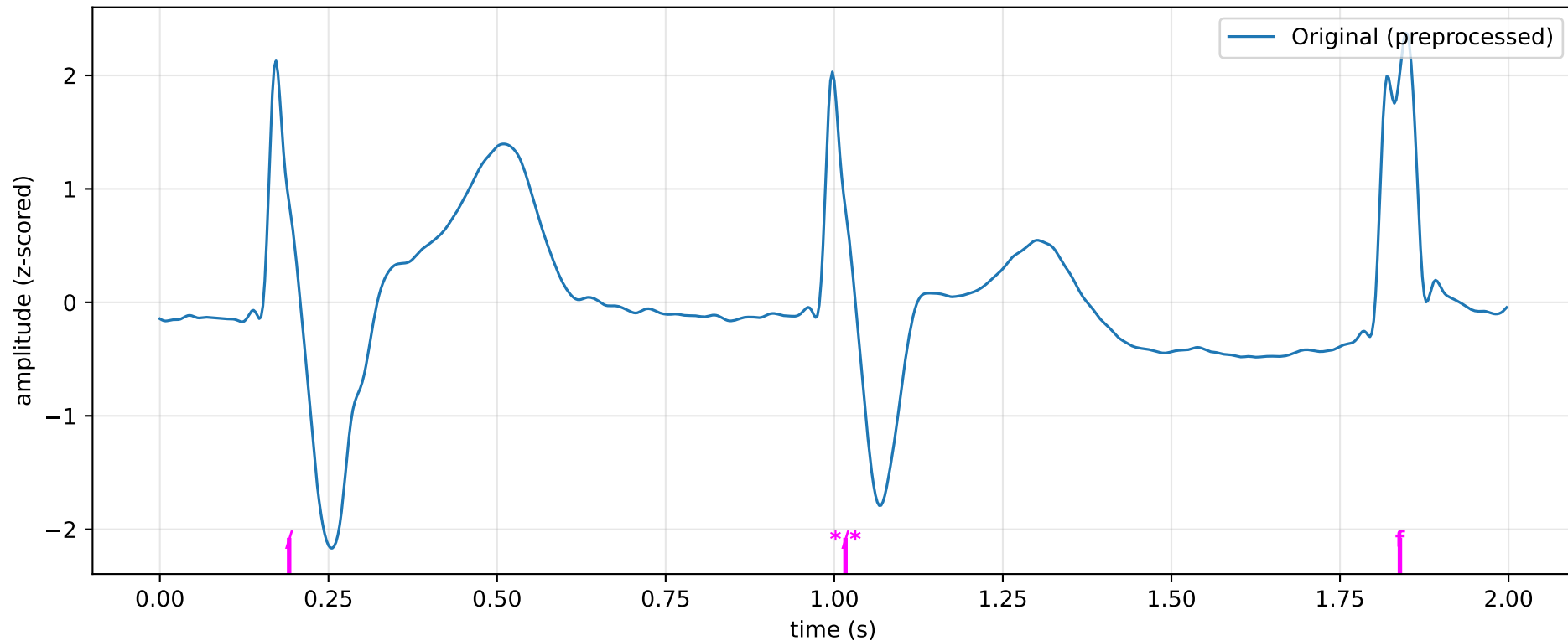


Symbol	R (centre beat - the one that labels the window)
Type	Right bundle branch block
Source	record 232, sample 630475
In window	+0ms:*R* +692ms:A
Residual	0.0433 Threshold 0.0434
Decision	NORMAL (truth: NORMAL)
Outcome	TN agree

NORMAL

Section: / - Paced beat

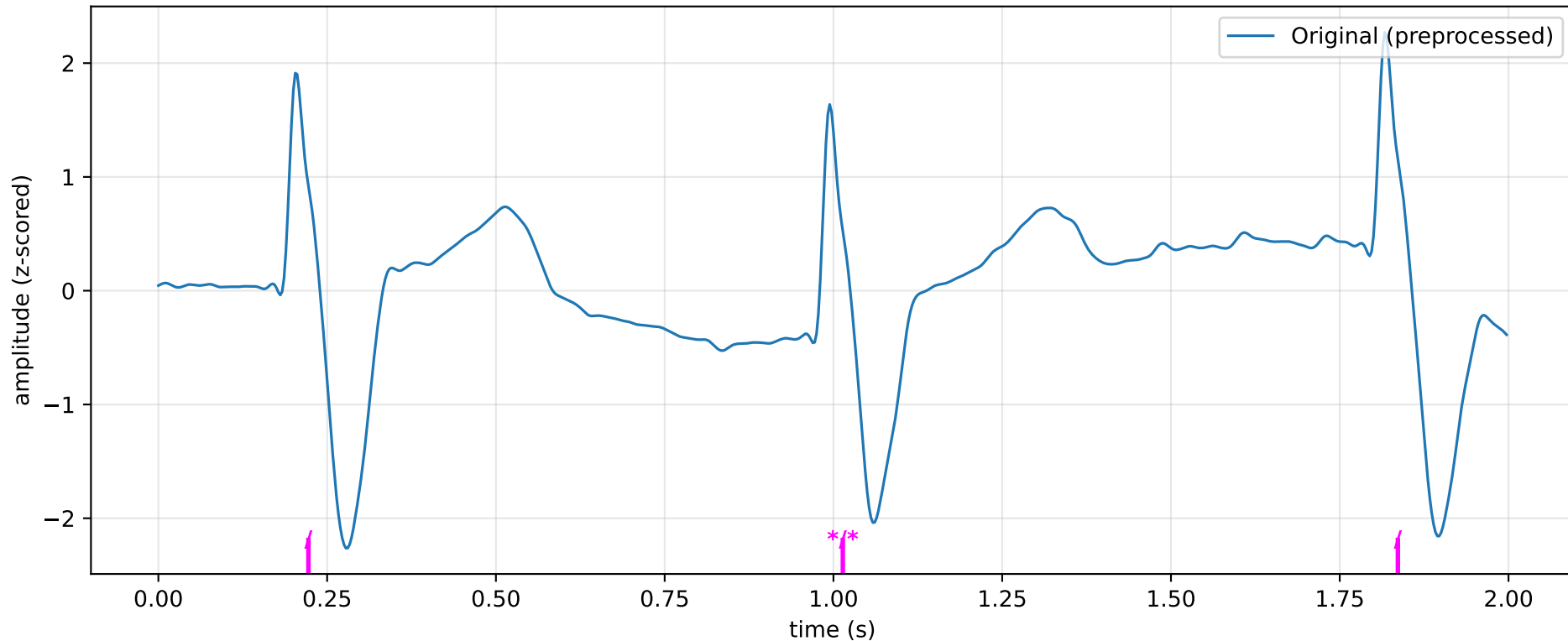
FN: Paced beat (/)



Symbol / (centre beat - the one that labels the window)
Type Paced beat
Source record 217, sample 28465
In window -808ms:/ +17ms:*/* +839ms:f
Residual 0.0429 Threshold 0.0434
Decision NORMAL (truth: ANOMALY)
Outcome FN mismatch

NORMAL

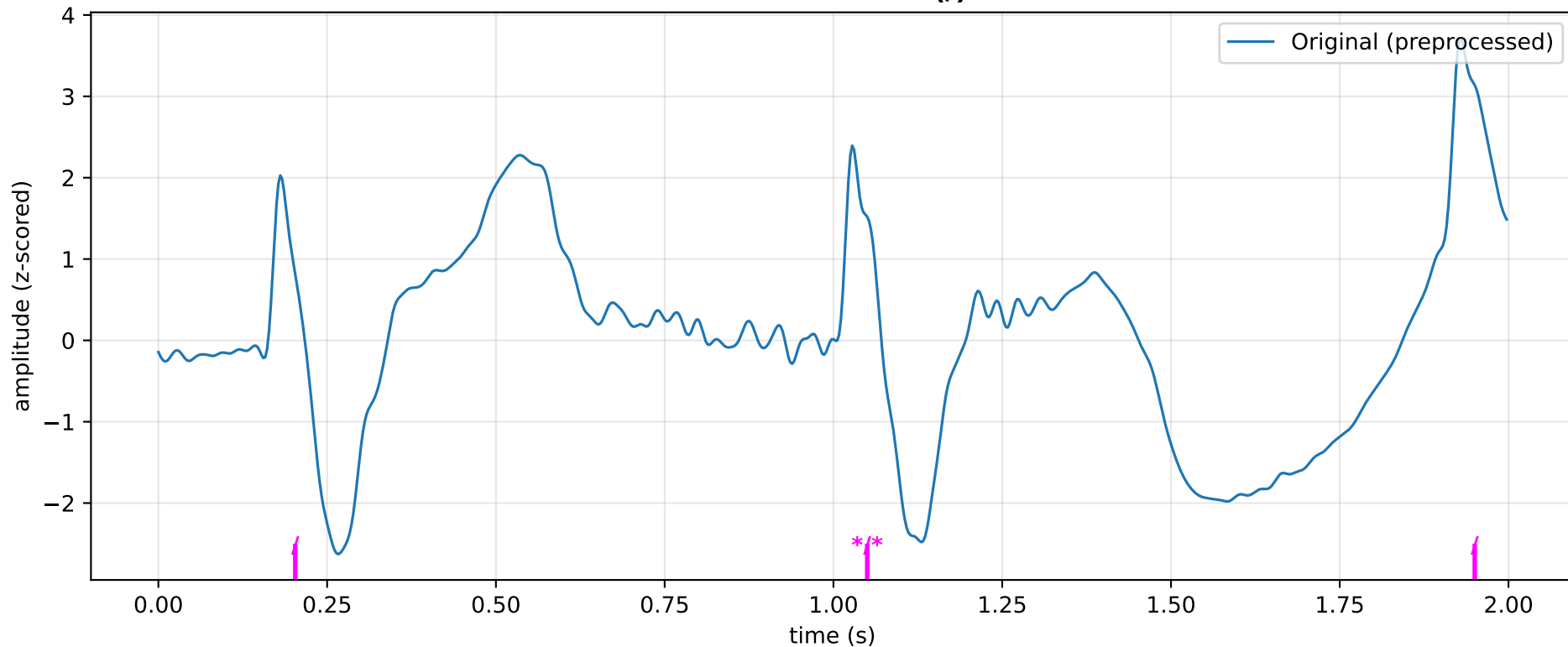
FN: Paced beat (/)



Symbol / (centre beat - the one that labels the window)
Type Paced beat
Source record 217, sample 602890
In window -778ms:/ +14ms:*/* +836ms:/
Residual 0.0371 Threshold 0.0434
Decision NORMAL (truth: ANOMALY)
Outcome FN mismatch

NORMAL

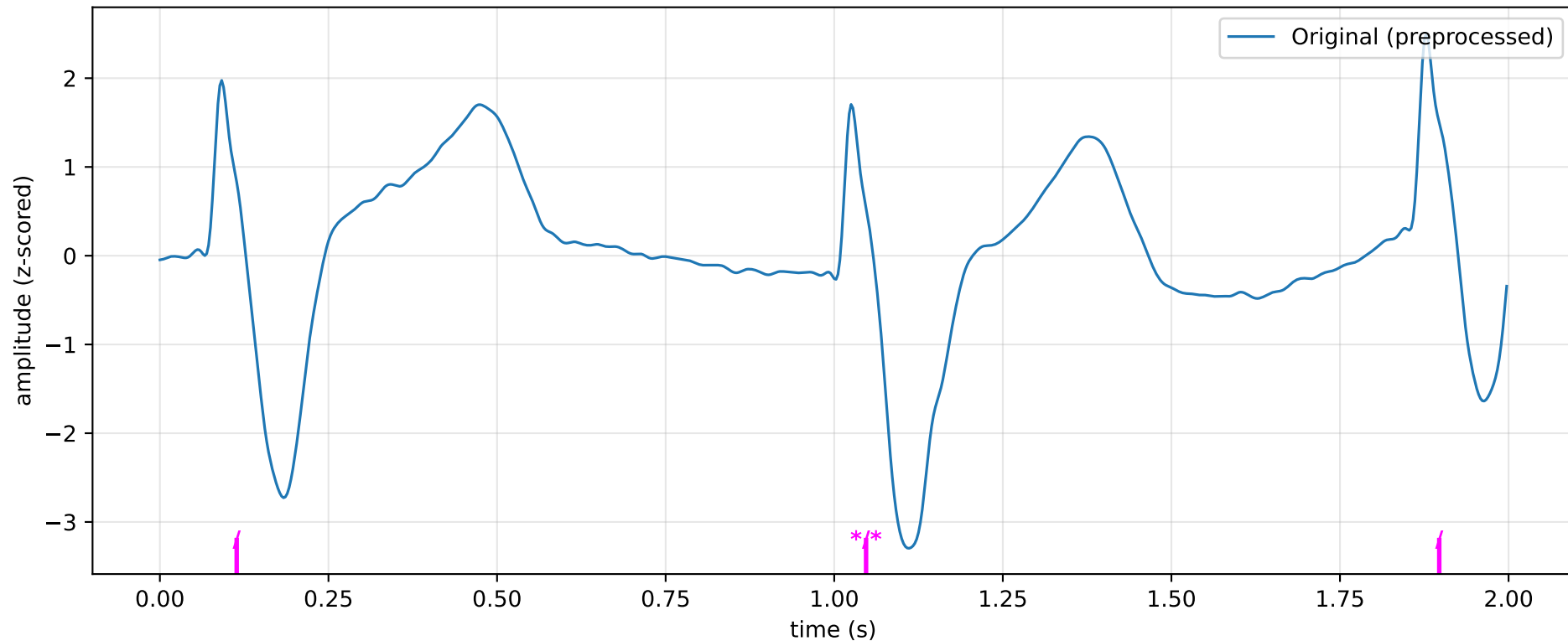
TP: Paced beat (/)



Symbol	/ (centre beat - the one that labels the window)
Type	Paced beat
Source	record 217, sample 482286
In window	-797ms:/ +50ms:*/* +950ms:/
Residual	0.2100 Threshold 0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP agree

ANOMALY

TP: Paced beat (/)

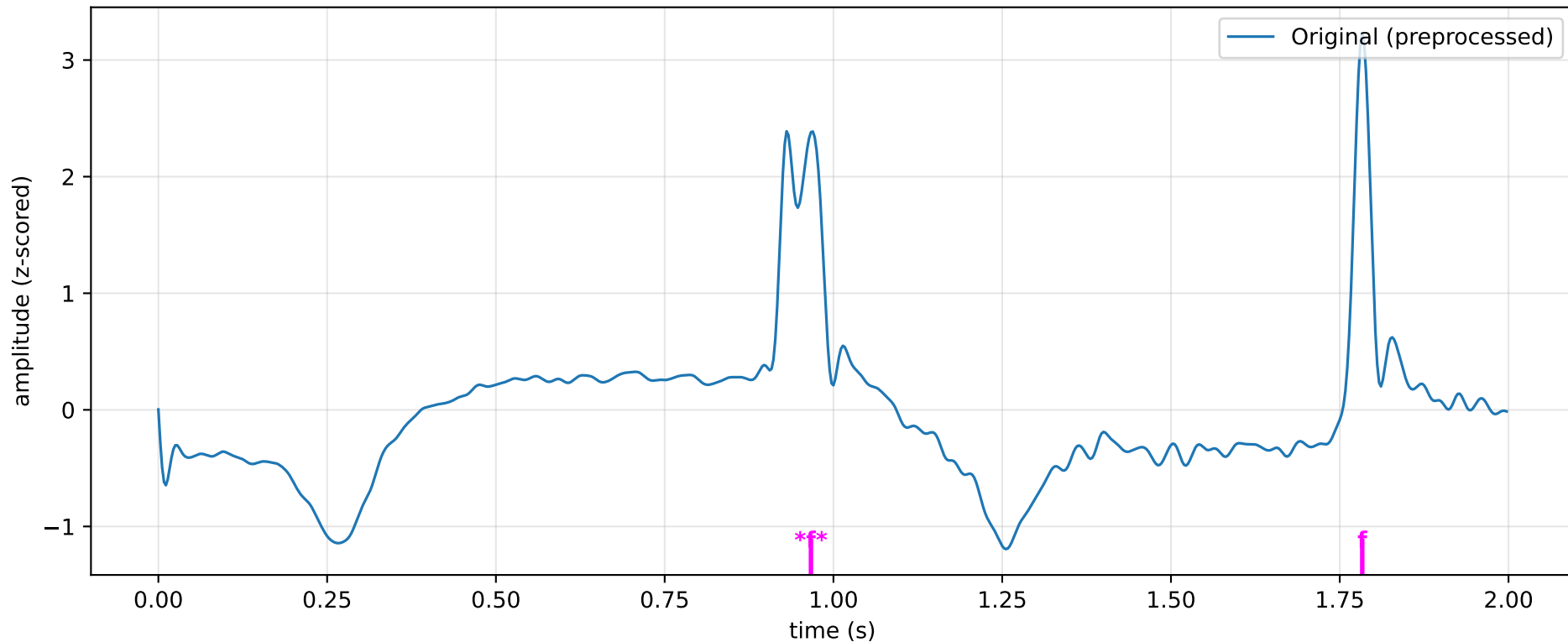


Symbol	/	(centre beat - the one that labels the window)
Type	Paced beat	
Source	record 217, sample 482947	
In window	-886ms:/	+47ms:*/* +897ms:/
Residual	0.1949	Threshold 0.0434
Decision	ANOMALY	(truth: ANOMALY)
Outcome	TP	agree

ANOMALY

Section: f - Fusion of paced and normal

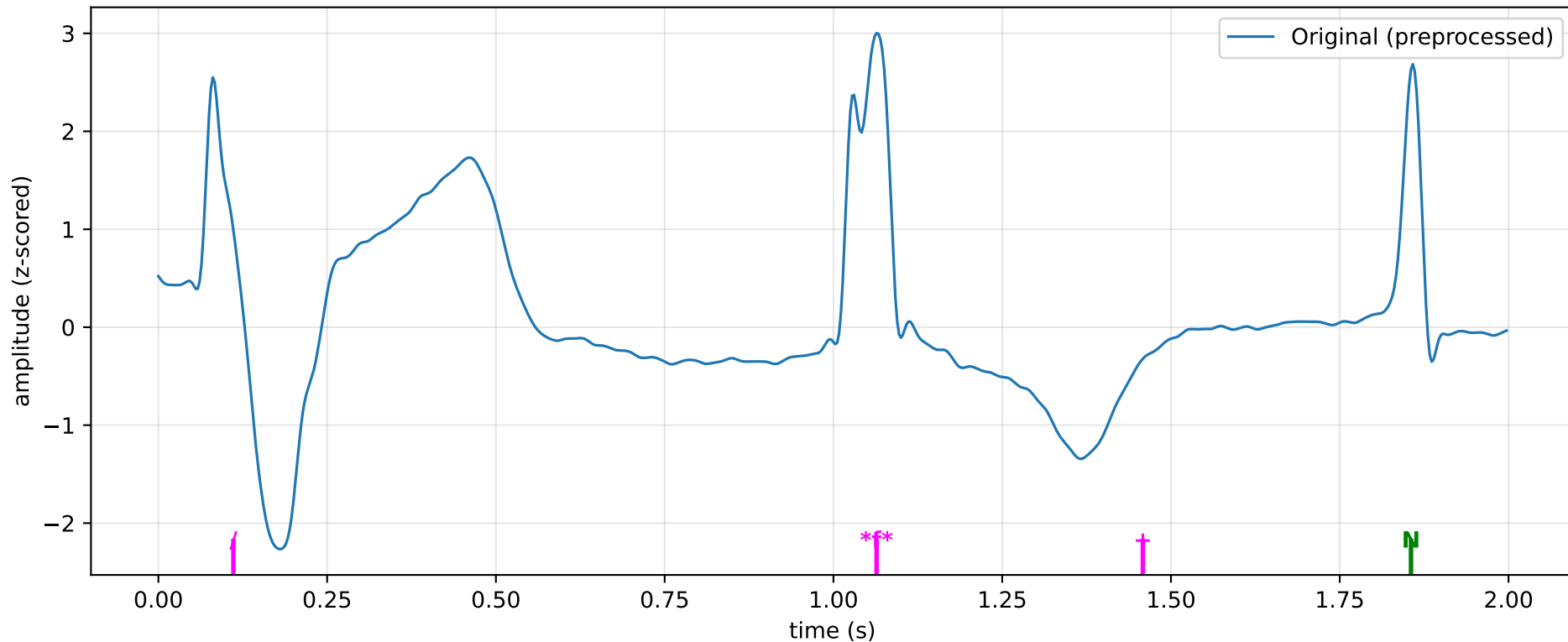
FN: Fusion of paced and normal (f)



Symbol	f (centre beat - the one that labels the window)
Type	Fusion of paced and normal
Source	record 217, sample 355577
In window	-33ms:*f* +783ms:f
Residual	0.0427 Threshold 0.0434
Decision	NORMAL (truth: ANOMALY)
Outcome	FN mismatch

NORMAL

TP: Fusion of paced and normal (f)



Symbol	f (centre beat - the one that labels the window)
Type	Fusion of paced and normal
Source	record 217, sample 316353
In window	-889ms:/ +64ms:*f* +458ms:+ +856ms:N
Residual	0.1961 Threshold 0.0434
Decision	ANOMALY (truth: ANOMALY)
Outcome	TP agree

ANOMALY